

MATHS EXTENSION 1**Calculus****Fully Worked Solutions****John Kinny-Lewis**

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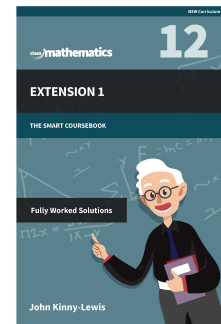
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Calculus

1

Integration by Substitution

Theory

To integrate $I = \int_1^3 x(x-1)^4 dx$ it would be tedious to expand and then integrate.

Instead, if we let $u = x - 1$ and also differentiate $u = x - 1$ to get $\frac{du}{dx} = 1$

Then I becomes $\int_{x=1}^{x=3} (u+1)u^4 du$ (since $du = dx$).

The limits also must be changed

$$\begin{aligned} \therefore I &= \int_0^2 u^5 + u^4 du \\ &= \left[\frac{1}{6}u^6 + \frac{1}{5}u^5 \right]_0^2 \\ &= \frac{1}{6} \times 2^6 + \frac{1}{5} \times 2^5 \\ &= \frac{256}{15}. \end{aligned}$$

Example 1 Using the substitution $u = x^2 + 2$, evaluate $\int_0^{\sqrt{2}} x\sqrt{x^2+2} dx$

Example 1 solution

$$I = \int_0^{\sqrt{2}} x\sqrt{x^2+2} dx$$

$$x = \sqrt{2} \rightarrow u = 4$$

$$x = 0 \rightarrow u = 2$$

$$u = x^2 + 2$$

$$\frac{du}{dx} = 2x$$

$$\frac{1}{2} du = x dx$$

Example cont

$$I = \frac{1}{2} \int_2^4 \sqrt{u} du$$

$$= \frac{1}{2} \int_2^4 u^{\frac{1}{2}} du$$

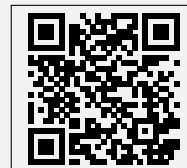
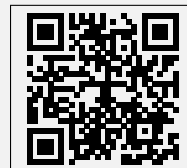
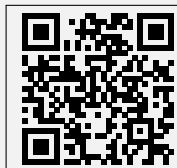
$$= \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_2^4$$

$$= \frac{1}{3} [4^{\frac{3}{2}} - 2^{\frac{3}{2}}]$$

$$= \frac{1}{3} [8 - 2\sqrt{2}]$$

$$= \frac{2}{3} [4 - \sqrt{2}]$$

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1. Using the substitution $x = \tan \theta$, evaluate $\int_0^1 \frac{1}{1+x^2} dx =$
2. Using the substitution $u = \sqrt{x}$, evaluate $\int_1^4 \frac{dx}{x + \sqrt{x}} =$
3. Using the substitution $u = 1 - x$, evaluate $\int \frac{x}{\sqrt{1-x}} dx =$
4. Using the substitution $u = 1 + x^2$, find $\int \frac{x}{\sqrt{1+x^2}} dx =$
5. Using the substitution $u = x^2$, find $\int x e^{x^2} dx =$
6. Using the substitution $u = t + 1$, evaluate $\int_0^3 t \sqrt{t+1} dt$
7. Using the substitution $u = \cos x$, evaluate $\int_0^{\frac{\pi}{3}} \frac{\sin x}{1 + \cos x} dx$
8. Using the substitution $u = \ln x$ find $\int \frac{1}{x \ln x} dx$
9. Using the substitution $u = e^x$, evaluate $I = \int_{-\ln 4}^{\ln 4} \frac{e^x}{1 + e^x} dx$
10. Using the substitution $u = e^x$, evaluate $I = \int_0^{\ln 3} \frac{1}{1 + e^{-x}} dx$
11. Using the substitution $u = x - 2$, evaluate the integral $\int_2^4 \frac{1}{x^2 - 4x + 8} dx$
12. Using the substitution $u = x^2 + 2$, evaluate $\int_0^{\sqrt{2}} x \sqrt{x^2 + 2} dx$
13. Find $\int 2x \cos(x^2 - 1) dx$, using $u = x^2 - 1$
14. Use the substitution $u^2 = 4 + x$, evaluate $\int_0^5 \frac{x}{\sqrt{4+x}} dx$ (Leave your answer in exact form.)
15. Using the substitution $u = x - 1$, evaluate $\int_1^2 \frac{1}{\sqrt{3+2x-x^2}} dx$
16.
 - i) Show that $\frac{u^2}{1+u^2} = 1 - \frac{1}{1+u^2}$
 - ii) Hence, by using the substitution $x = u^2$, evaluate the definite integral $\int_0^1 \frac{\sqrt{x}}{1+x} dx$

17. Using the substitution $x = 3 \sin \theta$

i) Show that $I = \int x\sqrt{9-x^2} dx$ can be written in the form $I = 27 \int \sin \theta \cos^2 \theta d\theta$

ii) Hence or otherwise evaluate the definite integral $\int_0^3 x\sqrt{9-x^2} dx$

18. i) Using the substitution $u = 1 - x^2$ show that $\int \frac{x}{\sqrt{1-x^2}} dx = -\sqrt{1-x^2} + C$

ii) Prove that $\sqrt{\frac{1-x}{1+x}} = \frac{1-x}{\sqrt{1-x^2}}$

iii) Using parts (i) and (ii), evaluate the integral $\int_0^{\frac{1}{2}} \sqrt{\frac{1-x}{1+x}} dx$

Past HSC Questions

19. HSC 2015 - Question 11e

20. HSC 2016 - Question 11b

21. HSC 2017 - Question 11e

22. HSC 2018 - Question 11f

23. HSC 2019 - Question 13a

24. HSC 2020 - Question 13a

25. HSC 2021 - Question 11c

26. HSC 2022 - Question 11b

27. HSC 2023 - Question 12a

Theory

Trigonometric Functions

When integrating trigonometry functions it is important to recall identities such as:

$$\cos 2\theta = 2\cos^2 \theta - 1$$

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$\sin \theta \cos \varphi = \frac{1}{2}(\sin(\theta + \varphi) + \sin(\theta - \varphi))$$

$$\cos \theta \cos \varphi = \frac{1}{2}(\cos(\theta + \varphi) + \cos(\theta - \varphi))$$

$$\cos \theta \sin \varphi = \frac{1}{2}(\sin(\theta + \varphi) - \sin(\theta - \varphi))$$

$$\sin \theta \sin \varphi = \frac{1}{2}(\cos(\theta - \varphi) - \cos(\theta + \varphi))$$

Example 1

Evaluate $\int_0^{\frac{\pi}{6}} \cos^2 \theta \, d\theta$

Example 1 solution

$$\begin{aligned} \int_0^{\frac{\pi}{6}} \cos^2 \theta \, d\theta &= \int_0^{\frac{\pi}{6}} \frac{1}{2}(\cos 2\theta + 1) \, d\theta \\ &= \frac{1}{2} \left[\frac{1}{2} \sin 2\theta + \theta \right]_0^{\frac{\pi}{6}} \\ &= \frac{1}{2} \left[\frac{1}{2} \sin \frac{\pi}{3} + \frac{\pi}{6} - 0 \right] \\ &= \frac{1}{4} \times \frac{\sqrt{3}}{2} + \frac{\pi}{12} \\ &= \frac{\sqrt{3}}{8} + \frac{\pi}{12} \end{aligned}$$

Example 2

Evaluate $\int_0^{\frac{\pi}{3}} \cos 3x \sin x \, dx$

Example 2 solution

$$\begin{aligned} \int_0^{\frac{\pi}{3}} \cos 3x \sin x \, dx &= \frac{1}{2} \int_0^{\frac{\pi}{3}} \sin(3x + x) - \sin(3x - x) \, dx \\ I &= \frac{1}{2} \int_0^{\frac{\pi}{3}} \sin 4x - \sin 2x \, dx \\ &= \frac{1}{2} \left[-\frac{1}{4} \cos 4x + \frac{1}{2} \cos 2x \right]_0^{\frac{\pi}{3}} \\ &= \frac{1}{2} \left[-\frac{1}{4} \cos \frac{4\pi}{3} + \frac{1}{2} \cos \frac{2\pi}{3} - \left(-\frac{1}{4} \cos 0 + \frac{1}{2} \cos 0 \right) \right] \\ &= \frac{1}{2} \left[-\frac{1}{4} \times -\frac{1}{2} + \frac{1}{2} \times -\frac{1}{2} + \frac{1}{4} - \frac{1}{2} \right] \\ &= \frac{1}{2} \left[\frac{1}{8} - \frac{1}{4} + \frac{1}{4} - \frac{1}{2} \right] \\ &= -\frac{3}{16} \end{aligned}$$

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28. Which expression is equal to $\int \sin^2 3x dx$?

$$A: \frac{1}{2} \left(x - \frac{1}{6} \sin 6x \right) + C \quad B: \frac{1}{2} \left(x + \frac{1}{6} \sin 6x \right) + C$$

$$C: \frac{1}{2} \left(x - \frac{1}{3} \sin 3x \right) + C \quad D: \frac{1}{2} \left(x + \frac{1}{3} \sin 3x \right) + C$$

29. $\int_0^{\frac{\pi}{3}} \sin^2 \theta d\theta =$

30. $\int_0^{\frac{\pi}{4}} \sin 2x \sin x dx =$

31. $\int \sin^3 x \cos x dx =$

32. $\int \tan^2 2x dx =$

33. $\int \tan x dx =$

34. $\int_{\pi/4}^{\pi/2} \sin x \cos^3 x dx =$

35. $\int_0^{\frac{\pi}{6}} \cos^2 \theta d\theta =$

36. $\int_0^{\frac{\pi}{3}} \cos 3x \sin x dx =$

37. $\int_{\frac{\pi}{8}}^{\frac{\pi}{4}} \sin^2 x \cos^2 x dx =$

38. $\int \frac{\cos x}{\sin^3 x} dx =$

39. Given that $f(x) = \frac{\cos x - \sin x}{\cos x + \sin x}$ find $f'(x)$ and hence evaluate $\int_0^{\frac{\pi}{4}} \frac{1}{(\sin x + \cos x)^2} dx$

40. i) Differentiate $\ln(\sec x + \tan x)$.

ii) Hence or otherwise evaluate the definite integral $\int_0^{\frac{\pi}{4}} \sec x dx$

41. i) Differentiate $\ln(\operatorname{cosec} x + \cot x)$

ii) Hence or otherwise evaluate the definite integral $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \operatorname{cosec} x dx$

42. Prove that $\cos(\theta + \phi) + \cos(\theta - \phi) = 2 \cos \theta \cos \phi$.

Hence evaluate $\int_0^{\frac{\pi}{3}} \cos \theta \cos 2\theta d\theta$

43. i) Show that $\sin(7x - 4x) + \sin(7x + 4x) = 2 \sin 7x \cos 4x$

ii) Hence find $\int \sin 7x \cos 4x dx$

Past HSC Questions

44. HSC 2015 - Question 11a

45. HSC 2016 - Question 5

46. HSC 2017 - Question 11f

47. HSC 2018 - Question 12a

48. HSC 2019 - Question 11e

49. HSC 2020 - Question 12d

50. HSC 2021 - Question 2

Theory

The following rules may be applied

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, -1 < x < 1$$

$$\frac{d}{dx}\left(\sin^{-1} \frac{x}{a}\right) = \frac{1}{\sqrt{a^2-x^2}}, -a < x < a$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}, -1 < x < 1$$

$$\frac{d}{dx}\left(\cos^{-1} \frac{x}{a}\right) = -\frac{1}{\sqrt{a^2-x^2}}, -a < x < a$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}, \text{ all } x$$

$$\frac{d}{dx}\left(\tan^{-1} \frac{x}{a}\right) = \frac{a}{a^2+x^2}, \text{ all } x$$

Example 1

Differentiate $y = \tan^{-1}\left(\frac{1}{x}\right)$

Example 1 solution

Let $u = \frac{1}{x} \rightarrow u = x^{-1}$

$$\frac{du}{dx} = -x^{-2} = -\frac{1}{x^2}$$

$$y = \tan^{-1} u$$

$$\frac{dy}{dx} = \frac{1}{1+u^2}$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{1}{1+u^2} \times -\frac{1}{x^2}$$

$$= \frac{1}{1+\frac{1}{x^2}} \times -\frac{1}{x^2}$$

$$= \frac{x^2}{1+x^2} \times -\frac{1}{x^2}$$

$$= \frac{-1}{1+x^2}$$

Example 2

Find the gradient of the tangent to $y = \sin^{-1} \frac{3x}{2}$
at $x = \frac{1}{3}$

Example 2 solution

$$y = \sin^{-1} \frac{3x}{2}$$

$$\text{Let } u = \frac{3x}{2} \quad y = \sin^{-1} u$$

$$\frac{du}{dx} = \frac{3}{2} \quad \frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

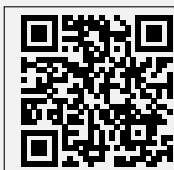
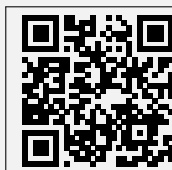
$$= \frac{1}{\sqrt{1-u^2}} \times \frac{3}{2}$$

$$= \frac{1}{\sqrt{1-\frac{9x^2}{4}}} \times \frac{3}{2}$$

$$= \frac{2}{\sqrt{4-9x^2}} \times \frac{3}{2} = \frac{3}{\sqrt{4-9x^2}}$$

$$\text{At } x = \frac{1}{3}, \frac{dy}{dx} = \frac{3}{\sqrt{4-1}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

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Differentiate

51. $y = \sin^{-1}(1-x)$

52. $y = \cos^{-1}\left(\frac{1}{x}\right)$

53. $y = \tan^{-1}\sqrt{x}$

54. $y = \ln(\tan^{-1}2x)$

55. $y = \tan^{-1}\left(\frac{1}{x}\right)$

56. Differentiate $y = x\sin^{-1}x$ and hence evaluate $\int_0^1 \sin^{-1}x \, dx$

57. Find the gradient of the tangent to $y = \sin^{-1}\frac{3x}{2}$ at $x = \frac{1}{3}$

58. Find the gradient of the tangent to $y = 2\sin^{-1}x$ at $x = -\frac{1}{2}$

59. Find the equation of the normal to the curve $y = 3\tan^{-1}x$ at $x = \sqrt{3}$

60. Consider the function $f(x) = \frac{1}{2}\tan^{-1}x$.

i) State the range of the function $y = f(x)$

ii) Sketch the graph of $y = f(x)$.

iii) Show that the equation of the tangent at $x = 1$ is $2x - 8y + \pi - 2 = 0$

61. The function $h(x)$ is given by $h(x) = \tan^{-1}x + \tan^{-1}\left(\frac{1}{x}\right)$

i) Find $h'(x)$

ii) Sketch the graph of $h(x)$

62. i) Differentiate $x\tan^{-1}x$

ii) Hence evaluate $\int_0^1 \tan^{-1}x \, dx$

63. i) Differentiate $x\sin^{-1}x + \sqrt{1-x^2}$.

ii) Hence, evaluate $\int_0^1 \sin^{-1}x \, dx$.

Past HSC Questions

64. HSC 2016 - Question 11c

65. HSC 2017 - Question 11b

66. HSC 2019 - Question 3

67. HSC 2020 - Question 13c

68. HSC 2021 - Question 14e

69. HSC 2022 - Question 12c

Theory

The following rules may be applied

$$\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + c$$

$$\int \frac{1}{\sqrt{a^2-x^2}} dx = \sin^{-1} \frac{x}{a} + c$$

$$\int \frac{-1}{\sqrt{1-x^2}} dx = \cos^{-1} x + c$$

$$\int \frac{-1}{\sqrt{a^2-x^2}} dx = \cos^{-1} \frac{x}{a} + c$$

$$\int \frac{1}{1+x^2} dx = \tan^{-1} x + c$$

$$\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + c.$$

Example 1

Evaluate $\int_{-1}^1 \frac{1}{\sqrt{2-x^2}} dx$

Example 1 solution

$$\begin{aligned} \int_{-1}^1 \frac{1}{\sqrt{2-x^2}} dx &= \int_{-1}^1 \frac{1}{\sqrt{(\sqrt{2})^2-x^2}} dx \\ &= \left[\sin^{-1} \frac{x}{\sqrt{2}} \right]_{-1}^1 \\ &= \left[\sin^{-1} \frac{1}{\sqrt{2}} - \sin^{-1} \frac{-1}{\sqrt{2}} \right] \\ &= \left[\sin^{-1} \frac{1}{\sqrt{2}} + \sin^{-1} \frac{1}{\sqrt{2}} \right] \\ &= 2 \sin^{-1} \frac{1}{\sqrt{2}} \\ &= 2 \times \frac{\pi}{4} \\ &= \frac{\pi}{2} \end{aligned}$$

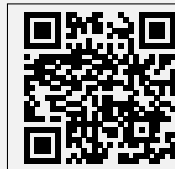
Example 2

Evaluate $\int_0^{\frac{1}{3}} \frac{1}{1+9x^2} dx$

Example 2 solution

$$\begin{aligned} \int_0^{\frac{1}{3}} \frac{1}{1+9x^2} dx &= \int_0^{\frac{1}{3}} \frac{1}{9\left(\frac{1}{9}+x^2\right)} dx \\ &= \frac{1}{9} \int_0^{\frac{1}{3}} \frac{1}{\left(\frac{1}{3}\right)^2+x^2} dx \\ &= \frac{1}{9} \left[3 \tan^{-1} 3x \right]_0^{\frac{1}{3}} \\ &= \frac{1}{3} [\tan^{-1}(1) - \tan^{-1}(0)] \\ &= \frac{1}{3} \times \frac{\pi}{4} \\ &= \frac{\pi}{12} \end{aligned}$$

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Evaluate the following:

$$70. \int_0^1 \frac{dx}{\sqrt{4-x^2}}$$

$$71. \int_0^{\frac{\sqrt{3}}{6}} \frac{-1}{\sqrt{1-9x^2}} dx$$

$$72. \int_0^{\frac{1}{2}} \frac{1}{1+4x^2} dx$$

$$73. \int_{-1}^1 \frac{1}{\sqrt{2-x^2}} dx$$

$$74. \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-2x^2}} dx$$

$$75. \int \frac{1}{\sqrt{1-4x^2}} dx =$$

$$76. \int_0^{\frac{1}{3}} \frac{1}{1+9x^2} dx$$

$$77. \int_0^{\frac{\sqrt{5}}{2}} \frac{dx}{\sqrt{5-4x^2}}$$

$$78. \int_0^2 \frac{1}{1+(x-1)^2} dx$$

79. What is the correct expression for $\int \frac{dx}{9+4x^2}$?

$$A : \frac{1}{4} \tan^{-1} \left(\frac{2x}{3} \right) \quad B : \frac{1}{3} \tan^{-1} \left(\frac{2x}{3} \right)$$

$$C : \frac{1}{6} \tan^{-1} \left(\frac{2x}{3} \right) \quad D : \frac{2}{3} \tan^{-1} \left(\frac{2x}{3} \right)$$

80. Find the exact value of $\int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx$

81. $\int_0^b \frac{dx}{b^2+x^2} = \frac{\pi}{3}$, find the value of b

82. Differentiate $x \cos^{-1} x - \sqrt{1-x^2}$ and hence evaluate $\int_0^1 \cos^{-1} x \, dx$

83. Find the area between the curve $y = \sin^{-1} x$, the x -axis and the line $x = \frac{1}{2}$

84. Sketch the curve $y = \sin^{-1} x$ and find the area between this curve, the x -axis and the lines $x = \frac{1}{2}$ and $x = 1$.

85. i) Using the substitution $u = \cos x$ show that $\int \tan x \, dx = -\ln(\cos x) + C$

ii) Neatly sketch the graph of $y = \tan^{-1} x$ and state its domain and range.

iii) Using part (i) and by considering the graph in part (ii) find the exact area of $\int_0^1 \tan^{-1} x \, dx$.

86. The function $g(x)$ is given by $g(x) = \sin^{-1} x + \cos^{-1} x$

i) Find $g'(x)$

ii) Hence or otherwise evaluate $\int_0^1 \sin^{-1} x + \cos^{-1} x \, dx$

Past HSC Questions

87. HSC 2015 - Question 7

88. HSC 2020 - Question 3

89. HSC 2021 - Question 11f

90. HSC 2023 - Question 11d

Theory

To determine the volume of a solid when a curve is rotated about the x -axis the formula

$$V = \pi \int_a^b y^2 dx \text{ is used}$$

If the curve is rotated about the y -axis the formula

$$V = \pi \int_a^b x^2 dy \text{ is used}$$

Example 1

Find the volume enclosed by the surface generated when the curve $x^2 + 3y^2 = 12$ is rotated about the x -axis.

Example 1 solution

Let $y = 0$ to find the x -intercepts,

$$x^2 = 12 \Rightarrow x = \pm\sqrt{12}$$

$$V = \pi \int y^2 dy$$

$$x^2 + 3y^2 = 12$$

$$\therefore y^2 = \frac{12 - x^2}{3}$$

$$V = \pi \int_{-\sqrt{12}}^{\sqrt{12}} \frac{12 - x^2}{3} dx$$

$$= \frac{2\pi}{3} \int_0^{\sqrt{12}} (12 - x^2) dx \quad (\text{even function})$$

$$= \frac{2\pi}{3} \left[12x - \frac{x^3}{3} \right]_0^{\sqrt{12}}$$

$$= \frac{2\pi}{3} \left[12\sqrt{12} - \frac{12\sqrt{12}}{3} \right]$$

$$= \frac{32\pi\sqrt{3}}{3} \text{ units}^3$$

Example 2

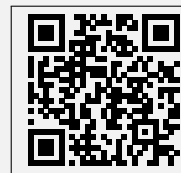
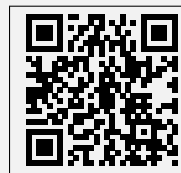
The region bounded by the parabola $y = x^2 - 2$ and the x -axis, is rotated about the y -axis. Find the volume of the solid generated.

Example 2 solution

$$y = x^2 - 2 \rightarrow x^2 = y + 2$$

$$\begin{aligned} V &= \pi \int_{-2}^0 y + 2 dy = \pi \left[\frac{y^2}{2} + 2y \right]_{-2}^0 \\ &= \pi[0 - (2 - 4)] \\ &= 2\pi \text{ u}^3 \end{aligned}$$

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91. The region bounded by the curve $y = e^{x/2}$ and the y -axis for $1 \leq y \leq 2$ is rotated about the y -axis to form a solid. Which of the following integrals would give the volume of the solid?

$$A : V = 4\pi \int_1^2 y \ln y \, dy \quad B : V = 4\pi \int_1^2 \ln y^2 \, dy$$

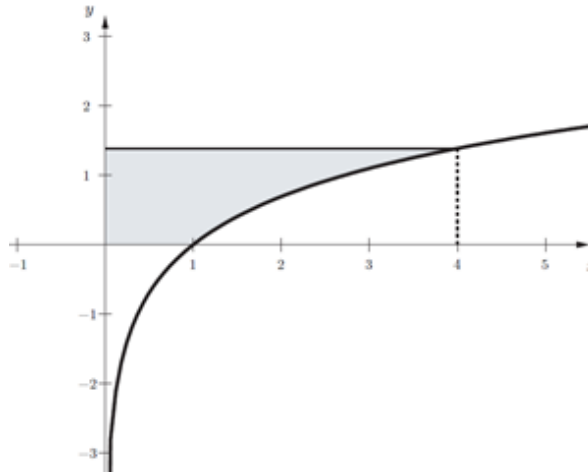
$$C : V = 4\pi \int_1^2 x e^x \, dx \quad D : V = 4\pi \int_1^2 (\ln y)^2 \, dy$$

92. The region between the curve $y = \frac{1}{x^2}$ and the x -axis from $x = 1$ to $x = 4$ is rotated about the x -axis. Calculate the exact volume of the solid formed.
93. The region between the curve $y = \frac{1}{x} + 1$ the y -axis and the lines $y = 2$ to $y = 4$ is rotated about the y -axis. Calculate the exact volume of the solid formed.
94. The region bounded by $y = e^{-x}$ the x -axis and the lines $x = 1$ and $x = 2$ is rotated about the x -axis. Calculate the exact volume of the solid formed.
95. The region bounded by the curve $y = \tan^{-1}x$ the y -axis and the lines $y = 0$ $y = \frac{\pi}{4}$ is rotated about the y -axis. Calculate the exact volume of the solid formed.
96. The line $y = -x + 2\frac{1}{2}$ intersects the hyperbola $y = \frac{1}{x}$ at $(\frac{1}{2}, 2)$ and $(2, \frac{1}{2})$. The region bounded by the line and the hyperbola is rotated around the x -axis. Calculate the exact volume of the solid formed.
97. The area under the curve $y = \sqrt{1 - \cos x}$ in the domain $-\pi \leq x \leq \pi$, is rotated about the x -axis. Find the volume of the solid of revolution obtained.
98. The region bounded by the parabola $y = x^2 - 2$ and the x -axis, is rotated about the y -axis. Find the volume of the solid generated.
99. The region bounded by $y = x^2$ and $y^2 = x$ is rotated about the x -axis. Calculate the volume of the solid generated.
100. Find the volume of the solid formed when $y = x^2 - 2$ is rotated about the y -axis from $y = 1$ to $y = 4$
101. Find the volume enclosed by the surface generated when the curve $x^2 + 3y^2 = 12$ is rotated about the x -axis.
102. i) Neatly sketch the graph of $y = \cos^{-1}x$ and state its domain and range.
- ii) Find the volume of the solid formed by rotating the curve $y = \cos^{-1}x$ between $y = 0$ and $y = \frac{\pi}{2}$ about the y -axis.

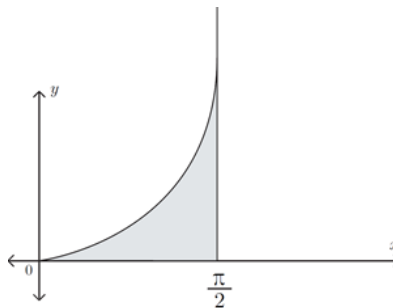
103. i) Show that $\int_0^{\frac{3\pi}{2}} \sin^2 x \, dx = \frac{3\pi}{4}$

- ii) The function $g(x)$ is given by $g(x) = 1 + \sin x$. The graph $y = g(x)$ for $0 \leq x \leq \frac{3\pi}{2}$ is rotated about the x axis. Find the volume generated (you may use the result in part i).

- 104.** The diagram shows the graph of $y = \ln x$. The shaded region, bounded by $y = \ln x$, the line $y = \ln 4$ and both the x and y axes, is rotated about the y -axis to form a solid.



- i) Show that the value of the solid is given by $V = \pi \int_0^{\ln 4} e^{2y} dy$
 - ii) Hence find the volume of the solid.
- 105.** The region beneath the curve $y = 1 - e^x$, which is above the x -axis and between the lines, $x = -1$ and $x = 0$ is rotated about the x -axis.
- i) Sketch the region.
 - ii) Find the volume of the resulting solid of revolution.
- 106.** The diagram shows part of the graph of the function $y = \tan \frac{x}{2}$. The curve, the x -axis and the line $\frac{\pi}{2}$ bound the shaded region. The region is rotated about the x -axis to form a solid.



- i) Show that the volume of the solid is given by $V = \pi \int_0^{\frac{\pi}{2}} \left(\sec^2 \frac{x}{2} - 1 \right) dx$
- ii) Find the exact volume of the solid.

107. HSC 2020 - Question 13b

108. HSC 2021 - Question 13a

109. HSC 2022 - Question 13b

110. HSC 2023 - Question 12e

Worked Solutions

Integration by Substitution

1. $x = \tan \theta$

$$\frac{dx}{d\theta} = \sec^2 \theta$$

$$dx = \sec^2 \theta d\theta$$

$$x = 1 \quad \tan \theta = 1$$

$$\theta = \frac{\pi}{4}$$

$$x = 0 \quad \tan \theta = 0$$

$$\theta = 0$$

$$\begin{aligned} \int_0^1 \frac{1}{1+x^2} dx &= \int_0^{\frac{\pi}{4}} \frac{1}{1+\tan^2 \theta} \times \sec^2 \theta d\theta \\ &= \int_0^{\frac{\pi}{4}} \frac{1}{\sec^2 \theta} \times \sec^2 \theta d\theta \\ &= \int_0^{\frac{\pi}{4}} 1 d\theta \\ &= [\theta]_0^{\frac{\pi}{4}} \\ &= \frac{\pi}{4} \end{aligned}$$

2. $u = x^{\frac{1}{2}}$

$$\frac{du}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$$

$$du = \frac{1}{2\sqrt{x}} dx$$

$$dx = 2u du$$

$$x = 4 \rightarrow u = 2$$

$$x = 1 \rightarrow u = 1$$

$$\begin{aligned} \int_1^4 \frac{dx}{x+\sqrt{x}} &= \int_1^2 \frac{2u du}{u^2+u} \\ &= \int_1^2 \frac{2 du}{u+1} \\ &= 2 [\log_e(u+1)]_1^2 \\ &= 2 [\log 3 - \log 2] \\ &= 2 \log \frac{3}{2} \end{aligned}$$

3. $u = 1 - x$

$$\frac{du}{dx} = -1$$

$$-du = dx$$

$$\begin{aligned} \int \frac{x}{\sqrt{1-x}} dx &= \int \frac{1-u}{\sqrt{u}} \times -du \\ &= \int \frac{u}{\sqrt{u}} - \frac{1}{\sqrt{u}} du \\ &= \int u^{\frac{1}{2}} - u^{-\frac{1}{2}} du \\ &= \frac{2}{3} u^{\frac{3}{2}} - 2u^{\frac{1}{2}} + c \\ &= \frac{2}{3} (1-x)^{3/2} - 2(1-x)^{\frac{1}{2}} \\ &= 2(1-x)^{\frac{1}{2}} \left\{ \frac{1-x}{3} - 1 \right\} \\ &= \frac{2}{3} \sqrt{1-x} (-x-2) \\ &= -\frac{2}{3} \sqrt{1-x} (x+2) + c \end{aligned}$$

4. $u = 1 + x^2$

$$\frac{du}{dx} = 2x \rightarrow \frac{1}{2} du = x dx$$

$$\begin{aligned} \int \frac{x}{\sqrt{1+x^2}} dx &= \frac{1}{2} \int \frac{du}{\sqrt{u}} = \frac{1}{2} \int u^{-\frac{1}{2}} du \\ &= \frac{1}{2} \times \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + c \\ &= \sqrt{1+x^2} + c \end{aligned}$$

5. Let $u = x^2$

$$\frac{du}{dx} = 2x \rightarrow \frac{1}{2} du = x dx$$

$$\begin{aligned} \int x e^{x^2} dx &= \int e^{x^2} x dx = \frac{1}{2} \int e^u du \\ &= \frac{1}{2} e^u + c \\ &= \frac{1}{2} e^{x^2} + c \end{aligned}$$

6. $u = t + 1 \quad t = 3 \rightarrow u = 4$

$$\frac{du}{dt} = 1 \quad t = 0 \rightarrow u = 1$$

$$du = dt$$

$$\begin{aligned} \int_0^3 t \sqrt{t+1} dt &= \int_1^4 (u-1) \sqrt{u} du \\ &= \int_1^4 u^{\frac{3}{2}} - u^{\frac{1}{2}} du \\ &= \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right]_1^4 \\ &= \left[\left(\frac{2}{5} \times 4^{\frac{5}{2}} - \frac{2}{3} 4^{\frac{3}{2}} \right) - \left(\frac{2}{5} - \frac{2}{3} \right) \right] \\ &= \left[\frac{64}{5} - \frac{16}{3} - \frac{2}{5} + \frac{2}{3} \right] = 7 \frac{11}{15} \end{aligned}$$

$$\begin{aligned}
7. \quad u &= \cos x & x &= \frac{\pi}{3} \rightarrow u = \cos \frac{\pi}{3} = \frac{1}{2} \\
\frac{du}{dx} &= -\sin x & x &= 0 \rightarrow u = \cos 0 = 1 \\
-du &= \sin x dx \\
\int_0^{\pi/3} \frac{\sin x}{1 + \cos x} dx &= - \int_1^{\frac{1}{2}} \frac{du}{1+u} \\
&= \int_{\frac{1}{2}}^1 \frac{du}{1+u} \\
&= [\log(1+u)]_{\frac{1}{2}}^1 \\
&= \log 2 - \log 1.5 = \log \frac{4}{3}
\end{aligned}$$

$$\begin{aligned}
8. \quad u &= \ln x \\
\frac{du}{dx} &= \frac{1}{x} \\
du &= \frac{dx}{x} \\
\int \frac{1}{x \ln x} dx &= \int \frac{1}{\ln x} \times \frac{dx}{x} \\
&= \int \frac{1}{u} du \\
&= \ln u + c \\
&= \ln(\ln x) + c
\end{aligned}$$

9. Firstly the limits,

$$\begin{aligned}
u &= e^x \\
u &= e^{\ln 4} \rightarrow u = 4 \\
u &= e^{-\ln 4} \\
&= e^{\ln \frac{1}{4}} \rightarrow u = \frac{1}{4} \\
u &= e^x \\
\therefore du &= e^x dx \\
\therefore I &= \int_{\frac{1}{4}}^4 \frac{du}{1+u} \\
I &= [\ln(1+u)]_{\frac{1}{4}}^4 \\
&= \ln 5 - \ln \frac{5}{4} \\
&= \ln 4
\end{aligned}$$

10. Firstly the limits,

$$\begin{aligned}
u &= e^x \\
u &= e^{\ln 3} \\
\therefore u &= 3 \text{ when } x = \ln 3 \\
u &= e^0 \\
\therefore u &= 1 \text{ when } x = 0 \\
u &= e^x \\
\frac{du}{dx} &= e^x \\
\therefore \frac{du}{dx} &= u \\
dx &= \frac{du}{u}
\end{aligned}$$

$$\begin{aligned}
\therefore I &= \int_1^3 \frac{1}{1 + \frac{1}{u}} \times \frac{du}{u} \\
&= \int_1^3 \frac{1}{u+1} du \\
&= [\ln(u+1)]_1^3 \\
&= \ln 4 - \ln 2 = \ln 2
\end{aligned}$$

11. $x = 4$ $u = 2$

$$\begin{aligned}
x &= 2 \quad u = 0 \\
I &= \int_0^2 \frac{1}{(u+2)^2 - 4(u+2) + 8} du \\
&= \int_0^2 \frac{1}{u^2 + 4u + 4 - 4u - 8 + 8} du \\
&= \int_0^2 \frac{1}{4 + u^2} du \\
&= \frac{1}{2} \left[\tan^{-1} \frac{u}{2} \right]_0^2 \\
&= \frac{1}{2} [\tan^{-1} 1 - \tan^{-1} 0] \\
&= \frac{\pi}{8}
\end{aligned}$$

12. $I = \int_0^{\sqrt{2}} x \sqrt{x^2 + 2} dx$

$$x = \sqrt{2} \rightarrow u = 4$$

$$x = 0 \rightarrow u = 2$$

$$u = x^2 + 2$$

$$\begin{aligned}
\frac{du}{dx} &= 2x \\
\frac{1}{2} du &= x dx \\
I &= \frac{1}{2} \int_2^4 \sqrt{u} du \\
&= \frac{1}{2} \int_2^4 u^{\frac{1}{2}} du \\
&= \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_2^4 \\
&= \frac{1}{3} [4^{\frac{3}{2}} - 2^{\frac{3}{2}}] \\
&= \frac{1}{3} [8 - 2\sqrt{2}] \\
&= \frac{2}{3} [4 - \sqrt{2}]
\end{aligned}$$

13. $I = \int 2x \cos(x^2 - 1) dx$

$$u = x^2 - 1$$

$$\frac{du}{dx} = 2x \rightarrow du = 2x dx$$

$$\begin{aligned}
\therefore I &= \int \cos u du \\
&= \sin u + c \\
&= \sin(x^2 - 1) + c
\end{aligned}$$

14. $I = \int_0^5 \frac{x}{\sqrt{4+x}} dx$

$$u^2 = 4 + x$$

$$u = (4 + x)^{\frac{1}{2}}$$

$$\frac{du}{dx} = \frac{1}{2}(4 + x)^{-\frac{1}{2}}$$

$$2du = \frac{1}{\sqrt{4 + x}} dx$$

$$x = u^2 - 4$$

$$\text{When } x = 5 \quad u = 3$$

$$x = 0 \quad u = 2$$

$$I = \int_2^3 (u^2 - 4) \times 2 du$$

$$= 2 \left[\frac{u^3}{3} - 4u \right]_2^3$$

$$= 2 \left[(9 - 12) - \left(\frac{8}{3} - 8 \right) \right]$$

$$= 4 \frac{2}{3}$$

15. Firstly the limits,

$$u = x - 1$$

$$u = 2 - 1 = 1$$

$$u = 1 - 1 = 0$$

$$u = x - 1$$

$$\therefore du = dx$$

$$I = \int_0^1 \frac{1}{\sqrt{3 + 2(u + 1) - (u + 1)^2}} du$$

$$= \int_0^1 \frac{1}{\sqrt{3 + 2u + 2 - u^2 - 2u - 1}} du$$

$$= \int_0^1 \frac{1}{\sqrt{4 - u^2}} du$$

$$= \left[\sin^{-1} \frac{u}{2} \right]_0^1 du$$

$$= \left[\sin^{-1} \frac{1}{2} - \sin^{-1} 0 \right] = \frac{\pi}{6}$$

16.

$$\text{i) } 1 - \frac{1}{1 + u^2} = \frac{1 + u^2 - 1}{1 + u^2} = \frac{u^2}{1 + u^2}$$

$$\text{ii) } I = \int_0^1 \frac{\sqrt{x}}{1 + x} dx$$

$$x = u^2$$

$$\frac{dx}{du} = 2u$$

$$dx = 2u du$$

$$\therefore I = \int_{x=0}^{x=1} \frac{u}{1 + u^2} \times 2u du$$

$$= 2 \int_{x=0}^{x=1} \frac{u^2}{1 + u^2} du$$

$$= 2 \int_{x=0}^{x=1} 1 - \frac{1}{1 + u^2} du \quad \text{from (i)}$$

$$= 2 \left[u - \tan^{-1} u \right]_{x=0}^{x=1}$$

$$= 2 \left[\sqrt{x} - \tan^{-1} \sqrt{x} \right]_0^1$$

$$= 2 \left[(1 - \tan^{-1} 1) - 0 \right] = 2 \left[1 - \frac{\pi}{4} \right] = 2 - \frac{\pi}{2}$$

17.

$$\text{i) } I = \int x \sqrt{9 - x^2} dx$$

$$x = 3 \sin \theta$$

$$\frac{dx}{d\theta} = 3 \cos \theta$$

$$\therefore dx = 3 \cos \theta d\theta$$

$$I = \int 3 \sin \theta \sqrt{9 - 9 \sin^2 \theta} \times 3 \cos \theta d\theta$$

$$= \int 9 \sin \theta \cos \theta \sqrt{9(1 - \sin^2 \theta)} d\theta$$

$$= 27 \int \sin \theta \cos \theta \sqrt{\cos^2 \theta} d\theta$$

$$= 27 \int \sin \theta \cos^2 \theta d\theta$$

ii) Finding the limits,

$$x = 3, \quad \therefore 3 \sin \theta = 3 \rightarrow \theta = \frac{\pi}{2}$$

$$x = 0, \quad \therefore 3 \sin \theta = 0 \rightarrow \theta = 0$$

$$\int_0^3 x \sqrt{9 - x^2} dx = 27 \int_0^{\frac{\pi}{2}} \sin \theta \cos^2 \theta d\theta$$

$$\text{Let } u = \cos \theta$$

$$du = -\sin \theta d\theta$$

$$= -27 \int_1^0 u^2 du$$

$$= -27 \left[\frac{u^3}{3} \right]_1^0$$

$$= -27 \left(0 - \frac{1}{3} \right) = 9$$

18.

i) Let $u = 1 - x^2$.

$$\frac{du}{dx} = -2x$$

$$\therefore x dx = -\frac{1}{2} du$$

$$\begin{aligned} I &= -\frac{1}{2} \int \frac{du}{\sqrt{u}} \\ &= -\frac{1}{2} \int u^{-\frac{1}{2}} du \\ &= -\frac{1}{2} \times \frac{2}{1} \times u^{\frac{1}{2}} + C \\ \therefore I &= -\sqrt{1-x^2} + C \end{aligned}$$

$$\begin{aligned} \text{ii)} \quad \sqrt{\frac{1-x}{1+x}} &= \sqrt{\frac{1-x}{1+x} \times \frac{1-x}{1-x}} \\ &= \frac{\sqrt{(1-x)^2}}{\sqrt{1-x^2}} \\ &= \frac{1-x}{\sqrt{1-x^2}} \\ \therefore \sqrt{\frac{1-x}{1+x}} &= \frac{1-x}{\sqrt{1-x^2}} \end{aligned}$$

$$\begin{aligned} \text{iii)} \quad \int_0^{\frac{1}{2}} \sqrt{\frac{1-x}{1+x}} dx &= \int_0^{\frac{1}{2}} \frac{1-x}{\sqrt{1-x^2}} dx \\ &= \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}} dx \\ &= \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}} dx + \int_0^{\frac{1}{2}} \frac{-x}{\sqrt{1-x^2}} dx \\ &= [\sin^{-1} x]_0^{\frac{1}{2}} + \left[\sqrt{1-x^2} \right]_0^{\frac{1}{2}} \\ &= \left[\sin^{-1} \left(\frac{1}{2} \right) - 0 \right] + \left[\sqrt{1-\frac{1}{4}} - \sqrt{1} \right] \\ &= \frac{\pi}{6} + \frac{\sqrt{3}}{2} - 1 \end{aligned}$$

19. Let $u = 2x - 1 \rightarrow \frac{du}{dx} = 2 \rightarrow dx = \frac{1}{2} du$

$$2x = u + 1 \rightarrow x = \frac{u+1}{2}$$

For $x = 2 \rightarrow u = 3$

$$x = 1 \rightarrow u = 1$$

$$\begin{aligned} I &= \int_1^3 \frac{1}{2} (u+1) \times \frac{1}{u^2} \times \frac{1}{2} du \\ &= \frac{1}{4} \int_1^3 \frac{u+1}{u^2} du \\ &= \frac{1}{4} \int_1^3 \frac{1}{u} + u^{-2} du \\ &= \frac{1}{4} \left[\ln u - \frac{1}{u} \right]_1^3 \\ I &= \frac{1}{4} \left[\ln 3 - \frac{1}{3} - (\ln 1 - 1) \right] \\ &= \frac{1}{4} \left[\ln 3 + \frac{2}{3} \right] = \frac{1}{12} [3 \ln 3 + 2] \end{aligned}$$

20. Let $u = x - 4 \rightarrow x = u + 4 \rightarrow \frac{du}{dx} = 1 \rightarrow dx = du$

$$\begin{aligned} I &= \int (u+4) u^{\frac{1}{2}} du \\ &= \int u^{\frac{3}{2}} + 4u^{\frac{1}{2}} du \\ &= \frac{2}{5} u^{\frac{5}{2}} + \frac{8}{3} u^{\frac{3}{2}} + C \\ &= \frac{2}{5} (x-4)^{\frac{5}{2}} + \frac{8}{3} (x-4)^{\frac{3}{2}} + C \end{aligned}$$

21. $x = u^2 - 1$

$$\frac{dx}{du} = 2u \rightarrow dx = 2u du$$

$$x+1 = u^2$$

$$\sqrt{x+1} = u$$

when $x = 3 \rightarrow u = 2$

$$x = 0 \rightarrow u = 1$$

$$\begin{aligned} I &= \int_1^2 \frac{u^2-1}{u} \times 2u du \\ &= 2 \int_1^2 (u - \frac{1}{u}) du \\ &= 2 \left[\frac{u^2}{2} - \ln u \right]_1^2 \\ &= 2 \left[\left(\frac{4}{2} - \ln 2 \right) - \left(\frac{1}{2} - \ln 1 \right) \right] = \frac{8}{3} \end{aligned}$$

22. $u = 1 - x \rightarrow x = 1 - u$

$$\frac{du}{dx} = -1 \quad \text{when } x = 0 \quad u = 1$$

$$dx = -du \quad x = -3 \quad u = 4$$

$$\begin{aligned} I &= \int_4^1 \frac{1-u}{\sqrt{u}} \times -du \\ &= \int_1^4 u^{-\frac{1}{2}} - u^{\frac{1}{2}} du \\ &= \left[2u^{\frac{1}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right]_1^4 \\ &= \left[\left(4 - \frac{16}{3} \right) - \left(2 - \frac{2}{3} \right) \right] \\ &= -\frac{2}{3} \end{aligned}$$

23. Let $u = \cos^2 x$

$$\frac{du}{dx} = -2 \cos x \sin x$$

$$du = -\sin 2x dx$$

$$\text{when } x = \frac{\pi}{4} \quad u = \frac{1}{2}$$

$$x = 0 \quad u = 1$$

$$\begin{aligned} I &= \int_1^{\frac{1}{2}} \frac{-du}{4+u} = \int_{\frac{1}{2}}^1 \frac{du}{u+4} = [\ln(u+4)]_{\frac{1}{2}}^1 \\ &= \ln 5 - \ln 4\frac{1}{2} \\ &= \ln(5 \div 4\frac{1}{2}) \\ &= \ln\left(\frac{10}{9}\right) \end{aligned}$$

24.

$$\text{i) } \frac{d}{d\theta} (\sin^3 \theta) = 3 \sin^2 \theta \times \cos \theta$$

$$\text{ii) } I = \int_0^1 \frac{x^2}{(1+x^2)^{\frac{5}{2}}} dx$$

$$x = 1 \quad \tan \theta = 1 \rightarrow \theta = \frac{\pi}{4}$$

$$x = 0 \quad \tan \theta = 0 \rightarrow \theta = 0$$

$$x = \tan \theta$$

$$\frac{dx}{d\theta} = \sec^2 \theta$$

$$I = \int_0^{\frac{\pi}{4}} \frac{\tan^2 \theta}{(1 + \tan^2 \theta)^{5/2}} \times \sec^2 \theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{\tan^2 \theta}{(\sec^2 \theta)^{5/2}} \times \sec^2 \theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} \tan^2 \theta \times \cos^3 \theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin^2 \theta}{\cos^2 \theta} \times \cos^3 \theta d\theta$$

$$I = \int_0^{\frac{\pi}{4}} \sin^2 \theta \cos \theta d\theta$$

$$= \frac{1}{3} [\sin^3 \theta]_0^{\frac{\pi}{4}} = \frac{1}{3} \left[\left(\frac{1}{\sqrt{2}} \right)^3 - 0 \right] = \frac{\sqrt{2}}{12}$$

$$25. I = \int x \sqrt{x+1} dx$$

$$u = x+1 \rightarrow \frac{du}{dx} = 1$$

$$I = \int (u-1)u^{\frac{1}{2}} du$$

$$= \int u^{\frac{3}{2}} - u^{\frac{1}{2}} du$$

$$= \frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} + c$$

$$= \frac{2}{5} (x+1)^{\frac{5}{2}} - \frac{2}{3} (x+1)^{\frac{3}{2}} + c$$

$$26. I = \int_0^1 \frac{x}{\sqrt{x^2+4}} dx$$

$$u = x^2 + 4$$

$$\frac{du}{dx} = 2x$$

$$\frac{1}{2} du = x dx$$

$$x = 1 \rightarrow u = 5$$

$$x = 0 \rightarrow u = 4$$

$$I = \frac{1}{2} \int_4^5 \frac{1}{\sqrt{u}} du$$

$$= \frac{1}{2} \int_4^5 u^{-\frac{1}{2}} du$$

$$= \frac{1}{2} \left[2u^{\frac{1}{2}} \right]_4^5 = \sqrt{5} - 2$$

$$27. I = \int_3^4 (x+2)\sqrt{x-3} dx$$

$$\text{Let } u = x-3 \quad x = u+3$$

$$\text{when } x = 4 \quad u = 1 \quad x+2 = u+5$$

$$x = 3 \quad u = 0$$

$$du = dx$$

$$I = \int_0^1 (u+5)\sqrt{u} du.$$

$$= \int_0^1 u^{\frac{3}{2}} + 5u^{\frac{1}{2}} du$$

$$= \left[\frac{2}{5} u^{\frac{5}{2}} + \frac{10}{3} u^{\frac{3}{2}} \right]_0^1$$

$$= \frac{2}{5} + \frac{10}{3} - 0$$

$$= \frac{56}{15}$$

Trig Integration

$$28. I = \int \sin^2 3x dx$$

$$\text{Let } u = 3x$$

$$\frac{du}{dx} = 3 \rightarrow dx = \frac{1}{3} du$$

$$I = \frac{1}{3} \int \sin^2 u du$$

$$= \frac{1}{3} \int \frac{1 - \cos 2u}{2} du$$

$$= \frac{1}{6} \left[u - \frac{1}{2} \sin 2u \right] + c$$

$$= \frac{1}{6} \left[3x - \frac{1}{2} \sin 6x \right] + c$$

$$= \frac{x}{2} - \frac{1}{12} \sin 6x + c$$

$$= \frac{1}{2} \left[x - \frac{1}{6} \sin 6x \right] + c \quad \therefore A$$

$$29. \int_0^{\frac{\pi}{3}} \sin^2 \theta d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{3}} 1 - \cos 2\theta d\theta \quad \left[\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta) \right]$$

$$= \frac{1}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{3}}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{3} - \frac{1}{2} \sin \frac{2\pi}{3} \right) - 0 \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right] = \frac{\pi}{6} - \frac{\sqrt{3}}{8}$$

$$30. \sin 2x \sin x = \frac{1}{2} (\cos(2x-x) - \cos(2x+x))$$

$$= \frac{1}{2} (\cos x - \cos 3x)$$

$$\begin{aligned}\frac{1}{2} \int_0^{\pi/4} \cos x - \cos 3x dx &= \frac{1}{2} \left[\sin x - \frac{1}{3} \sin 3x \right]_0^{\pi/4} \\ &= \frac{1}{2} \left[\sin \frac{\pi}{4} - \frac{1}{3} \sin \frac{3\pi}{4} - 0 \right] \\ &= \frac{1}{2} \left[\frac{\sqrt{2}}{2} - \frac{1}{3} \frac{\sqrt{2}}{2} \right] \\ &= \frac{\sqrt{2}}{4} \left[1 - \frac{1}{3} \right] = \frac{\sqrt{2}}{6}\end{aligned}$$

$$\begin{aligned}31. \quad \int [f(x)]^n f'(x) dx &= \frac{1}{n+1} [f(x)]^{n+1} + C \\ \therefore \int (\sin x)^3 \cos x dx &= \frac{1}{4} \sin^4 x + C\end{aligned}$$

$$\begin{aligned}32. \quad \int \tan^2 2x dx &= \int \sec^2 2x - 1 dx \\ &= \frac{1}{2} \tan 2x - x + c\end{aligned}$$

$$\begin{aligned}33. \quad \int \frac{f'(x)}{f(x)} dx &= \log_e f(x) + c \\ \int \tan x dx &= \int \frac{\sin x}{\cos x} dx = - \int \frac{-\sin x}{\cos x} dx \\ &= -\log_e |\cos x| + c\end{aligned}$$

$$\begin{aligned}34. \quad I &= - \int_{\pi/4}^{\pi/2} -\sin x (\cos x)^3 dx \\ &= - \left[\frac{(\cos x)^4}{4} \right]_{\pi/4}^{\pi/2} \\ I &= \frac{1}{4} [(\cos x)^4]_{\pi/2}^{\pi/4} \\ &= \frac{1}{4} \left[\left(\frac{1}{\sqrt{2}} \right)^4 - 0 \right] = \frac{1}{16}\end{aligned}$$

$$\begin{aligned}35. \quad \int_0^{\pi/6} \cos^2 \theta d\theta &= \int_0^{\pi/6} \frac{1}{2} (\cos 2\theta + 1) d\theta \\ &= \frac{1}{2} \left[\frac{1}{2} \sin 2\theta + \theta \right]_0^{\pi/6} \\ &= \frac{1}{2} \left[\frac{1}{2} \sin \frac{\pi}{3} + \frac{\pi}{6} - 0 \right] \\ &= \frac{1}{4} \times \frac{\sqrt{3}}{2} + \frac{\pi}{12} \\ &= \frac{\sqrt{3}}{8} + \frac{\pi}{12}\end{aligned}$$

$$36. \quad \int_0^{\pi/3} \cos 3x \sin x dx = \frac{1}{2} \int_0^{\pi/3} \sin(3x+x) - \sin(3x-x) dx$$

$$\begin{aligned}I &= \frac{1}{2} \int_0^{\pi/3} \sin 4x - \sin 2x dx \\ &= \frac{1}{2} \left[-\frac{1}{4} \cos 4x + \frac{1}{2} \cos 2x \right]_0^{\pi/3} \\ &= \frac{1}{2} \left[-\frac{1}{4} \cos \frac{4\pi}{3} + \frac{1}{2} \cos \frac{2\pi}{3} - \left(-\frac{1}{4} \cos 0 + \frac{1}{2} \cos 0 \right) \right] \\ &= \frac{1}{2} \left[-\frac{1}{4} \times -\frac{1}{2} + \frac{1}{2} \times -\frac{1}{2} + \frac{1}{4} - \frac{1}{2} \right] \\ &= \frac{1}{2} \left[\frac{1}{8} - \frac{1}{4} + \frac{1}{4} - \frac{1}{2} \right] \\ &= -\frac{3}{16}\end{aligned}$$

$$\begin{aligned}37. \quad \sin 2x &= 2 \sin x \cos x \\ \sin^2 2x &= 4 \sin^2 x \cos^2 x\end{aligned}$$

$$\therefore \sin^2 x \cos^2 x = \frac{1}{4} \sin^2 2x$$

$$\begin{aligned}\therefore I &= \frac{1}{4} \int_{\pi/8}^{\pi/4} \sin^2 2x dx \\ &= \frac{1}{4} \int_{\pi/8}^{\pi/4} \frac{1}{2} (1 - \cos 4x) dx\end{aligned}$$

$$\begin{aligned}\therefore I &= \frac{1}{8} \int_{\pi/8}^{\pi/4} 1 - \cos 4x dx \\ &= \frac{1}{8} \left[x - \frac{1}{4} \sin 4x \right]_{\pi/8}^{\pi/4} \\ &= \frac{1}{8} \left[\left(\frac{\pi}{4} - 0 \right) - \left(\frac{\pi}{8} - \frac{1}{4} \right) \right] \\ &= \frac{1}{8} \left[\frac{\pi}{8} + \frac{1}{4} \right] \\ &= \frac{1}{64} [\pi + 2]\end{aligned}$$

$$\begin{aligned}38. \quad \int \frac{\cos x}{\sin^3 x} dx &= \int (\sin x)^{-3} \cos x dx \\ &= \frac{(\sin x)^{-2}}{-2} + c \\ &= -\frac{1}{2} \times \frac{1}{\sin^2 x} + c \\ &= -\frac{1}{2} \operatorname{cosec}^2 x + c\end{aligned}$$

39. Let $u = \cos x - \sin x$ and $v = \cos x + \sin x$

$$\therefore \frac{du}{dx} = -\sin x - \cos x, \frac{dv}{dx} = -\sin x + \cos x$$

$$\begin{aligned} f'(x) &= \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \\ &= \frac{-(\cos x + \sin x)(\cos x + \sin x) - (\cos x - \sin x)(\cos x - \sin x)}{(\cos x + \sin x)^2} \\ &= \frac{-(\cos^2 x + 2\sin x \cos x + \sin^2 x) - (\cos^2 x - 2\sin x \cos x + \sin^2 x)}{(\cos x + \sin x)^2} \\ &= \frac{-(1 + 2\sin x \cos x) - (1 - 2\sin x \cos x)}{(\cos x + \sin x)^2} \\ &= \frac{-2}{(\cos x + \sin x)^2} \\ \therefore \int \frac{-2}{(\cos x + \sin x)^2} dx &= \frac{\cos x - \sin x}{\cos x + \sin x} + C \\ \therefore \int_0^{\frac{\pi}{4}} \frac{1}{(\sin x + \cos x)^2} dx &= -\frac{1}{2} \left[\frac{\cos x - \sin x}{\cos x + \sin x} \right]_0^{\frac{\pi}{4}} \\ &= -\frac{1}{2} \left[0 - \frac{1}{1} \right] \\ &= \frac{1}{2} \end{aligned}$$

40.

$$\begin{aligned} \text{i) } \frac{d}{dx} [\ln(\sec x + \tan x)] &= \frac{\sec x \tan x + \sec^2 x}{\sec x + \tan x} \\ &= \frac{\sec x(\tan x + \sec x)}{\sec x + \tan x} \\ &= \sec x \end{aligned}$$

$$\begin{aligned} \text{ii) } \int_0^{\frac{\pi}{4}} \sec x dx &= [\ln(\sec x + \tan x)]_0^{\frac{\pi}{4}} \\ &= [\ln(\sqrt{2} + 1) - \ln(1 + 0)] \\ &= \ln(\sqrt{2} + 1) \end{aligned}$$

41.

$$\begin{aligned} \text{i) Let } y &= \ln(\operatorname{cosec} x + \cot x) \\ \therefore \frac{dy}{dx} &= \frac{-\operatorname{cosec} x \cot x - \operatorname{cosec}^2 x}{\operatorname{cosec} x + \cot x} \\ &= \frac{-\operatorname{cosec} x(\cot x + \operatorname{cosec} x)}{\cot x + \operatorname{cosec} x} \\ &= -\operatorname{cosec} x \end{aligned}$$

$$\begin{aligned} \text{ii) } \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \operatorname{cosec} x dx &= -[\ln(\operatorname{cosec} x + \cot x)]_{\frac{\pi}{6}}^{\frac{\pi}{3}} \\ &= [\ln(\operatorname{cosec} x + \cot x)]_{\frac{\pi}{3}}^{\frac{\pi}{6}} \\ &= \left[\ln\left(2 + \sqrt{3}\right) - \ln\left(\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}\right) \right] \\ &= \ln(2 + \sqrt{3}) - \ln \sqrt{3} \\ &= \ln \frac{2 + \sqrt{3}}{\sqrt{3}} \\ &= \ln \frac{3 + 2\sqrt{3}}{3} \end{aligned}$$

42. $\cos(\theta + \phi) + \cos(\theta - \phi)$

$$\begin{aligned} &= \cos \theta \cos \phi - \sin \theta \sin \phi + \cos \theta \cos \phi + \sin \theta \sin \phi \\ &= 2 \cos \theta \cos \phi \end{aligned}$$

$$\begin{aligned} \int_0^{\frac{\pi}{3}} \cos \theta \cos 2\theta d\theta &= \frac{1}{2} \int_0^{\frac{\pi}{3}} \cos 3\theta + \cos \theta d\theta \\ &= \frac{1}{2} \left[\frac{1}{3} \sin 3\theta + \sin \theta \right]_0^{\frac{\pi}{3}} \\ &= \frac{1}{2} \left[\left(\frac{1}{3} \sin \pi + \sin \frac{\pi}{3} \right) - 0 \right] \\ &= \frac{1}{2} \times \frac{\sqrt{3}}{2} \\ &= \frac{\sqrt{3}}{4} \end{aligned}$$

43.

$$\text{i) } 2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$\text{Let } A = 7x \text{ and } B = 4x$$

$$\text{LHS} = 2 \sin 7x \cos 4x$$

$$\text{RHS} = \sin(7x + 4x) + \sin(7x - 4x)$$

$$\therefore \text{LHS} = \text{RHS}$$

$$\begin{aligned} \text{ii) } \int \sin 7x \cos 4x dx &= \int \sin 11x + \sin 3x dx \\ &= -\frac{1}{11} \cos 11x - \frac{1}{3} \cos 3x + c \end{aligned}$$

$$44. \cos 2x = 1 - 2 \sin^2 x \rightarrow \sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\begin{aligned} \therefore \int \sin^2 x dx &= \frac{1}{2} \int 1 - \cos 2x dx \\ &= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right] + C \end{aligned}$$

$$45. \cos 4x = 1 - 2 \sin^2 2x \rightarrow \sin^2 2x = \frac{1}{2}(1 - \cos 4x)$$

$$\begin{aligned} I &= \frac{1}{2} \int 1 - \cos 4x dx \\ &= \frac{1}{2} \left[x - \frac{1}{4} \sin 4x \right] + C \quad \therefore A \end{aligned}$$

$$46. \int f'(x)[f(x)]^n dx = \frac{[-f(x)]^{n+1}}{n+1} + c$$

$$\therefore \int \cos x (\sin x)^2 dx = \frac{\sin^3 x}{3} + c$$

$$47. \cos 2x = 2 \cos^2 x - 1$$

$$\cos^2 x = \frac{\cos 2x + 1}{2} \rightarrow \cos^2(3x) = \frac{\cos 6x + 1}{2}$$

$$\begin{aligned} \therefore I &= \frac{1}{2} \int \cos 6x + 1 dx \\ &= \frac{1}{2} \left[\frac{1}{6} \sin 6x + x \right] + C \\ &= \frac{1}{12} \sin 6x + \frac{x}{2} + C \end{aligned}$$

$$48. \cos 2x = 1 - 2 \sin^2 x$$

$$2 \sin^2 x = 1 - \cos 2x$$

$$2 \sin^2 4x = 1 - \cos 8x$$

$$\begin{aligned} I &= \int 1 - \cos 8x dx \\ &= x - \frac{1}{8} \sin 8x + c \end{aligned}$$

$$\begin{aligned}
49. \quad & \int_0^{\frac{\pi}{2}} \cos 5x \sin 3x dx \\
&= \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin(5x+3x) - \sin(5x-3x) dx \\
&= \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin 8x - \sin 2x dx \\
&= \frac{1}{2} \left[-\frac{1}{8} \cos 8x + \frac{1}{2} \cos 2x \right]_0^{\pi/2} \\
&= \frac{1}{2} \left[-\frac{1}{8} \cos 4\pi + \frac{1}{2} \cos \pi - \left(-\frac{1}{8} \cos 0 + \frac{1}{2} \cos 0 \right) \right] \\
&= \frac{1}{2} \left[-\frac{1}{8} - \frac{1}{2} + \frac{1}{8} - \frac{1}{2} \right] \\
&= -\frac{1}{2}
\end{aligned}$$

$$\begin{aligned}
50. \quad & \cos 2x = 1 - 2\sin^2 x \\
& \sin^2 x = \frac{1}{2}(1 - \cos 2x) \\
& \therefore \int \sin^2 3x dx = \int \frac{1 - \cos 6x}{2} dx \quad \therefore B
\end{aligned}$$

Derivative of Inverse Functions

$$\begin{aligned}
51. \quad & y = \sin^{-1}(1-x) \\
& \text{Let } u = 1-x \quad y = \sin^{-1} u \\
& \frac{du}{dx} = -1 \quad \frac{dy}{du} = \frac{1}{\sqrt{1-u^2}} \\
& \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \\
&= \frac{-1}{\sqrt{1-u^2}} \\
& \frac{dy}{dx} = \frac{-1}{\sqrt{1-(1-x)^2}} \\
&= \frac{-1}{\sqrt{1-(1-2x+x^2)}} \\
&= \frac{-1}{\sqrt{2x-x^2}} \\
&= \frac{-1}{\sqrt{x(2-x)}}
\end{aligned}$$

$$\begin{aligned}
52. \quad & y = \cos^{-1}\left(\frac{1}{x}\right) \\
& \text{Let } u = \frac{1}{x} = x^{-1} \quad y = \cos^{-1} u \\
& \frac{du}{dx} = -x^{-2} = \frac{-1}{x^2} \quad \frac{dy}{du} = \frac{-1}{\sqrt{1-u^2}} \\
& \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \\
&= \frac{-1}{\sqrt{1-u^2}} \times \frac{-1}{x^2} \\
&= \frac{1}{x^2 \sqrt{1-\frac{1}{x^2}}} \\
&= \frac{1}{x \sqrt{x^2-1}}
\end{aligned}$$

$$\begin{aligned}
53. \quad & y = \tan^{-1} \sqrt{x} \\
& \text{Let } u = \sqrt{x} = x^{\frac{1}{2}} \quad y = \tan^{-1} u \\
& \frac{du}{dx} = \frac{1}{2} x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}} \quad \frac{dy}{du} = \frac{1}{1+u^2} \\
& \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \\
&= \frac{1}{1+u^2} \times \frac{1}{2\sqrt{x}} \\
&= \frac{1}{2\sqrt{x}(1+x)}
\end{aligned}$$

$$\begin{aligned}
54. \quad & \text{Let } u = \tan^{-1} 2x \\
& \frac{du}{dx} = \frac{1}{1+(2x)^2} \times 2 \quad y = \ln u \\
&= \frac{2}{1+4x^2} \quad \frac{dy}{du} = \frac{1}{u} \\
& \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \\
&= \frac{1}{u} \times \frac{2}{1+4x^2} \\
& \therefore \frac{dy}{dx} = \frac{2}{\tan^{-1} 2x (1+4x^2)}
\end{aligned}$$

$$\begin{aligned}
55. \quad & \text{Let } u = \frac{1}{x} \\
& u = x^{-1} \\
& \frac{du}{dx} = -x^{-2} \\
&= -\frac{1}{x^2} \\
& y = \tan^{-1} u \\
& \frac{dy}{du} = \frac{1}{1+u^2} \\
& \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \\
&= \frac{1}{1+u^2} \times -\frac{1}{x^2} \\
&= \frac{1}{1+\frac{1}{x^2}} \times -\frac{1}{x^2} \\
&= \frac{x^2}{1+x^2} \times -\frac{1}{x^2} = \frac{-1}{1+x^2}
\end{aligned}$$

$$\begin{aligned}
56. \quad & y = x \sin^{-1} x \\
& \frac{dy}{dx} = \sin^{-1} x + \frac{x}{\sqrt{1-x^2}} \\
& \int \sin^{-1} x + \frac{x}{\sqrt{1-x^2}} dx = x \sin^{-1} x + c \\
& \int_0^1 \sin^{-1} x dx = [x \sin^{-1} x]_0^1 - \int_0^1 \frac{x}{\sqrt{1-x^2}} dx \\
&= (1 \times \sin^{-1}(1) - 0) + \frac{1}{2} \int_0^1 -2x(1-x^2)^{-\frac{1}{2}} dx \\
&= \frac{\pi}{2} + \frac{1}{2} \left[2(1-x^2)^{\frac{1}{2}} \right]_0^1 \\
&= \frac{\pi}{2} + \frac{1}{2} [0 - 2] = \frac{\pi}{2} - 1
\end{aligned}$$

57. $y = \sin^{-1} \frac{3x}{2}$

Let $u = \frac{3x}{2}$ $y = \sin^{-1} u$

$$\frac{du}{dx} = \frac{3}{2} \quad \frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{1}{\sqrt{1-u^2}} \times \frac{3}{2}$$

$$= \frac{1}{\sqrt{1-\frac{9x^2}{4}}} \times \frac{3}{2}$$

$$= \frac{2}{\sqrt{4-9x^2}} \times \frac{3}{2} = \frac{3}{\sqrt{4-9x^2}}$$

At $x = \frac{1}{3}$, $\frac{dy}{dx} = \frac{3}{\sqrt{4-1}} = \frac{3}{\sqrt{3}} = \sqrt{3}$

58. $y = 2 \sin^{-1} x$

$$y' = 2 \times \frac{1}{\sqrt{1-x^2}}$$

At $x = -\frac{1}{2}$

$$y' = \frac{2}{\sqrt{1-\frac{1}{4}}} = \frac{2}{\sqrt{\frac{3}{4}}} = \frac{2}{\frac{\sqrt{3}}{2}}$$

$$= \frac{4}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{4\sqrt{3}}{3}$$

59. $y = 3 \tan^{-1} x$

$$y' = \frac{3}{1+x^2}$$

At $x = \sqrt{3}$

$$y = 3 \tan^{-1} \sqrt{3}$$

$$= 3 \times \frac{\pi}{3} = \pi$$

$$y' = \frac{3}{1+3} = \frac{3}{4}$$

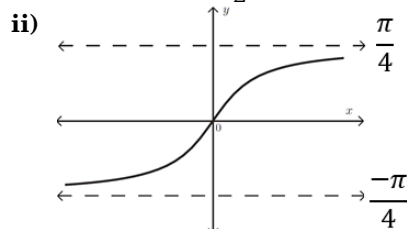
$$\therefore y - \pi = -\frac{4}{3}(x - \sqrt{3})$$

$$3y - 3\pi = -4x + 4\sqrt{3}$$

$$4x + 3y - 3\pi - 4\sqrt{3} = 0$$

60.

- i) The range of $f(x) = \tan^{-1} x$ is $-\frac{\pi}{2} < f(x) < \frac{\pi}{2}$, hence the range of $f(x) = \frac{1}{2} \tan^{-1} x$ is $-\frac{\pi}{4} < f(x) < \frac{\pi}{4}$



$$y = \frac{1}{2} \tan^{-1} x$$

iii) $\frac{dy}{dx} = \frac{1}{2} \times \frac{1}{1+x^2}$

$$= \frac{1}{2(1+x^2)}$$

at $x = 1$ $\frac{dy}{dx} = \frac{1}{4}$ $y = \frac{1}{2} \tan^{-1}(1) = \frac{\pi}{8}$

$$\therefore y - \frac{\pi}{8} = \frac{1}{4}(x - 1)$$

$$8y - \pi = 2x - 2$$

$$2x - 8y + \pi - 2 = 0$$

61.

i) $h(x) = \tan^{-1} x + \tan^{-1} \left(\frac{1}{x} \right)$

$$h'(x) = \frac{1}{1+x^2} + \frac{-1}{x^2} \times \frac{1}{1+\left(\frac{1}{x}\right)^2}$$

$$= \frac{1}{1+x^2} - \frac{1}{x^2} \times \frac{x^2}{1+x^2}$$

$$= \frac{1}{1+x^2} - \frac{1}{1+x^2}$$

$$= 0$$

- ii) If $h'(x) = 0$, then $h(x) = K$ for some constant K .

Let $\tan^{-1} x + \tan^{-1} \left(\frac{1}{x} \right) = K$,

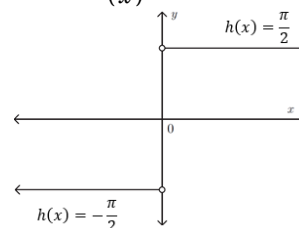
Let $x = 1$, $\tan^{-1}(1) + \tan^{-1}(1) = \frac{\pi}{2}$

$$\therefore K = \frac{\pi}{2} \text{ (for positive } x)$$

Let $x = -1$, $\tan^{-1}(-1) + \tan^{-1}(-1) = \frac{-\pi}{2}$

$$\therefore K = \frac{-\pi}{2} \text{ (for negative } x)$$

At $x = 0$, $\tan^{-1} \left(\frac{1}{x} \right)$ is not defined.



62.

i) $\frac{d}{dx} (x \tan^{-1} x) = \tan^{-1} x + x \times \frac{1}{1+x^2}$

$$= \tan^{-1} x + \frac{x}{1+x^2}$$

ii) $\int_0^1 \tan^{-1} x dx + \int_0^1 \frac{x}{1+x^2} dx = [x \tan^{-1} x]_0^1$

$$\int_0^1 \tan^{-1} x dx = [x \tan^{-1} x]_0^1 - \frac{1}{2} \int_0^1 \frac{2x}{1+x^2} dx$$

$$= \frac{\pi}{4} - \frac{1}{2} [\ln(1+x^2)]_0^1$$

$$= \frac{\pi}{4} - \frac{1}{2} [\ln 2 - \ln 1]$$

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

63.

i) Let $y = x \sin^{-1} x + (1 - x^2)^{\frac{1}{2}}$

$$\begin{aligned}\frac{dy}{dx} &= x \times \frac{1}{\sqrt{1-x^2}} + \sin^{-1} x + \frac{1}{2} (1-x^2)^{-\frac{1}{2}} \times -2x \\ &= \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x - \frac{x}{\sqrt{1-x^2}} \\ &= \sin^{-1} x\end{aligned}$$

$$\begin{aligned}\text{ii) } \int_0^1 \sin^{-1} x \, dx &= \left[x \sin^{-1} x + \sqrt{1-x^2} \right]_0^1 \\ &= \left[\sin^{-1}(1) + 0 - (0 + \sqrt{1}) \right] \\ &= \frac{\pi}{2} - 1\end{aligned}$$

$$\begin{aligned}64. \quad \frac{d}{dx} (3 \tan^{-1}(2x)) &= 3 \times \frac{1}{1+(2x)^2} \times 2 \\ &= \frac{6}{1+4x^2}\end{aligned}$$

$$\begin{aligned}65. \quad \frac{d}{dx} (\tan^{-1}(x^3)) &= \frac{1}{1+(x^3)^2} \times 3x^2 \\ &= \frac{3x^2}{1+x^6}\end{aligned}$$

$$\begin{aligned}66. \quad \text{Let } y &= \tan^{-1} \left(\frac{1}{2}x \right) \\ y' &= \frac{1}{1+\left(\frac{x}{2}\right)^2} \times \frac{1}{2} \\ &= \frac{1}{1+\frac{x^2}{4}} \times \frac{1}{2} \\ &= \frac{4}{4+x^2} \times \frac{1}{2} \\ &= \frac{2}{4+x^2} \quad \therefore \text{C.}\end{aligned}$$

67.

i) $f(x) = \tan(\cos^{-1}(x))$

Let $y = \tan(\cos^{-1}(x))$

Let $u = \cos^{-1}(x)$ $y = \tan u$

$$\begin{aligned}\frac{du}{dx} &= \frac{-1}{\sqrt{1-x^2}} \quad \frac{dy}{du} = \sec^2 u. \\ \therefore \frac{dy}{dx} &= \sec^2 u \times \frac{-1}{\sqrt{1-x^2}} \\ &= \frac{1}{\cos^2 u} \times \frac{-1}{\sqrt{1-x^2}} \\ &= \frac{1}{(\cos(\cos^{-1} x))^2} \times \frac{-1}{\sqrt{1-x^2}} \\ \therefore f'(x) &= \frac{-1}{x^2 \sqrt{1-x^2}}\end{aligned}$$

$$\begin{aligned}g(x) &= \frac{(1-x^2)^{\frac{1}{2}}}{x} \\ g'(x) &= \frac{x \times \frac{1}{2} (1-x^2)^{-\frac{1}{2}} \times -2x - (1-x^2)^{\frac{1}{2}}}{x^2} \\ &= \frac{\frac{-x^2}{\sqrt{1-x^2}} - \sqrt{1-x^2}}{x^2} \\ &= \frac{-x^2 - 1 + x^2}{x^2 \sqrt{1-x^2}} \\ &= \frac{-1}{x^2 \sqrt{1-x^2}}\end{aligned}$$

$$\therefore f'(x) = g'(x)$$

ii) $f'(x) = g'(x)$

$$\therefore f'(x) - g'(x) = 0$$

$$\therefore f(x) - g(x) = k \quad (k \text{ a constant})$$

For $x < 0 \rightarrow x = -1$

$$\begin{aligned}f(-1) - g(-1) &= \tan(\cos^{-1}(-1)) - \frac{\sqrt{1-(-1)^2}}{-1} \\ &= \tan \pi + 0 \\ &= 0\end{aligned}$$

For $x > 0 \rightarrow x = 1$

$$\begin{aligned}f(1) - g(1) &= \tan(\cos^{-1}(1)) - \frac{\sqrt{1-(1)^2}}{1} \\ &= \tan(0) - 0 \\ &= 0\end{aligned}$$

So for all x $f(x) = g(x)$ in the domain $-1 \leq x \leq 1$

$$68. \quad g(x) = x^3 + 4x - 2$$

let $g(x) = y$

$$y = x^3 + 4x - 2$$

$\therefore$ Inverse is : $x = y^3 + 4y - 2$

$$\frac{dx}{dy} = 3y^2 + 4$$

$$\therefore \frac{dy}{dx} = \frac{1}{3y^2 + 4} \left(\frac{d}{dx} (g^{-1}x) \right)$$

$$f(x) = xg^{-1}(x)$$

$$\begin{aligned}f'(x) &= g^{-1}(x) + x \times \frac{d}{dx} (g^{-1}x) \\ &= g^{-1}(3) + 3 \times \frac{1}{3(g^{-1}(3))^2 + 4} \\ &= 1 + \frac{3}{7} \\ &= \frac{10}{7}\end{aligned}$$

$$\begin{aligned}
 69. \quad y &= x \tan^{-1} x \\
 u &= x \quad v = \tan^{-1} x \\
 du &= 1 \quad dv = \frac{1}{1+x^2} \\
 y' &= \frac{x}{1+x^2} + \tan^{-1} x \\
 \text{At } x &= 1 \quad y' = \frac{1}{2} + \frac{\pi}{4} \\
 \left(1, \frac{\pi}{4}\right) \quad y - \frac{\pi}{4} &= \left(\frac{1}{2} + \frac{\pi}{4}\right)(x-1) \\
 y - \frac{\pi}{4} &= \left(\frac{1}{2} + \frac{\pi}{4}\right)x - \frac{1}{2} - \frac{\pi}{4} \\
 \therefore y &= \left(\frac{1}{2} + \frac{\pi}{4}\right)x - \frac{1}{2}
 \end{aligned}$$

Integrations that give Inverse Trig Functions

$$\begin{aligned}
 70. \quad \int_0^1 \frac{dx}{\sqrt{4-x^2}} &= \int_0^1 \frac{dx}{\sqrt{2^2-x^2}} = \left[\sin^{-1} \frac{x}{2} \right]_0^1 \\
 &= \left[\sin^{-1} \frac{1}{2} - \sin^{-1} 0 \right] = \frac{\pi}{6}
 \end{aligned}$$

$$\begin{aligned}
 71. \quad \int_0^{\frac{\sqrt{3}}{6}} \frac{-1}{\sqrt{1-9x^2}} dx &= \int_0^{\frac{\sqrt{3}}{6}} \frac{-1}{\sqrt{9\left(\frac{1}{9}-x^2\right)}} dx \\
 &= \frac{1}{3} \int_0^{\frac{\sqrt{3}}{6}} \frac{-1}{\sqrt{\left(\frac{1}{3}\right)^2-x^2}} dx \\
 &= \frac{1}{3} \left[\cos^{-1} 3x \right]_0^{\frac{\sqrt{3}}{6}} \\
 &= \frac{1}{3} \left[\cos^{-1} \frac{\sqrt{3}}{2} - \cos^{-1} 0 \right] \\
 &= \frac{1}{3} \left[\frac{\pi}{6} - \frac{\pi}{2} \right] \\
 &= -\frac{\pi}{9}
 \end{aligned}$$

$$\begin{aligned}
 72. \quad \int_0^{\frac{1}{2}} \frac{1}{1+4x^2} dx &= \int_0^{\frac{1}{2}} \frac{1}{4\left(\frac{1}{4}+x^2\right)} dx \\
 &= \frac{1}{4} \int_0^{\frac{1}{2}} \frac{1}{\left(\frac{1}{2}\right)^2+x^2} dx \\
 &= \frac{1}{4} \left[2 \tan^{-1} 2x \right]_0^{\frac{1}{2}} \\
 &= \frac{1}{2} \left[\tan^{-1} 1 - \tan^{-1} 0 \right] \\
 &= \frac{\pi}{8}
 \end{aligned}$$

$$\begin{aligned}
 73. \quad \int_{-1}^1 \frac{1}{\sqrt{2-x^2}} dx &= \int_{-1}^1 \frac{1}{\sqrt{(\sqrt{2})^2-x^2}} dx \\
 &= \left[\sin^{-1} \frac{x}{\sqrt{2}} \right]_{-1}^1 \\
 &= \left[\sin^{-1} \frac{1}{\sqrt{2}} - \sin^{-1} \frac{-1}{\sqrt{2}} \right] \\
 &= \left[\sin^{-1} \frac{1}{\sqrt{2}} + \sin^{-1} \frac{1}{\sqrt{2}} \right] \\
 &= 2 \sin^{-1} \frac{1}{\sqrt{2}} \\
 &= 2 \times \frac{\pi}{4} = \frac{\pi}{2}
 \end{aligned}$$

$$\begin{aligned}
 74. \quad I &= \int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-2x^2}} dx = \int_0^{\frac{1}{2}} \frac{1}{\sqrt{2\left(\frac{1}{2}-x^2\right)}} dx \\
 &= \frac{1}{\sqrt{2}} \int_0^{\frac{1}{2}} \frac{1}{\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2-x^2}} dx \\
 &= \frac{1}{\sqrt{2}} \left[\sin^{-1} \sqrt{2}x \right]_0^{\frac{1}{2}} \\
 &= \frac{1}{\sqrt{2}} \left[\sin^{-1} \frac{\sqrt{2}}{2} - \sin^{-1} 0 \right] \\
 &= \frac{1}{\sqrt{2}} \times \frac{\pi}{4} \\
 &= \frac{\pi\sqrt{2}}{8}
 \end{aligned}$$

$$\begin{aligned}
 75. \quad \int \frac{1}{\sqrt{1-4x^2}} dx &= \int \frac{1}{\sqrt{4\left(\frac{1}{4}-x^2\right)}} dx \\
 &= \frac{1}{2} \int \frac{1}{\sqrt{\left(\frac{1}{2}\right)^2-x^2}} dx \\
 &= \frac{1}{2} \sin^{-1} 2x + c
 \end{aligned}$$

$$\begin{aligned}
 76. \quad \int_0^{\frac{1}{3}} \frac{1}{1+9x^2} dx &= \int_0^{\frac{1}{3}} \frac{1}{9\left(\frac{1}{9}+x^2\right)} dx \\
 &= \frac{1}{9} \int_0^{\frac{1}{3}} \frac{1}{\left(\frac{1}{3}\right)^2+x^2} dx \\
 &= \frac{1}{9} \left[3 \tan^{-1} 3x \right]_0^{\frac{1}{3}} \\
 &= \frac{1}{3} \left[\tan^{-1}(1) - \tan^{-1}(0) \right] \\
 &= \frac{1}{3} \times \frac{\pi}{4} = \frac{\pi}{12}
 \end{aligned}$$

$$\begin{aligned}
 77. \int_0^{\frac{\sqrt{5}}{2}} \frac{dx}{\sqrt{5-4x^2}} &= \int_0^{\frac{\sqrt{5}}{2}} \frac{dx}{\sqrt{4\left(\frac{5}{4}-x^2\right)}} \\
 &= \frac{1}{2} \int_0^{\frac{\sqrt{5}}{2}} \frac{dx}{\sqrt{\left(\frac{\sqrt{5}}{2}\right)^2 - x^2}} \\
 &= \frac{1}{2} \left[\sin^{-1} \frac{2x}{\sqrt{5}} \right]_0^{\frac{\sqrt{5}}{2}} \\
 &= \frac{1}{2} [\sin^{-1}(1) - \sin^{-1}(0)] \\
 &= \frac{1}{2} \left[\frac{\pi}{2} - 0 \right] \\
 &= \frac{\pi}{4}
 \end{aligned}$$

$$\begin{aligned}
 78. \int_0^2 \frac{1}{1+(x-1)^2} dx &= [\tan^{-1}(x-1)]_0^2 \\
 &= [\tan^{-1}(1) - \tan^{-1}(-1)] \\
 &= [\tan^{-1}(1) + \tan^{-1}(1)] \\
 &= \frac{\pi}{4} + \frac{\pi}{4} \\
 &= \frac{\pi}{2}
 \end{aligned}$$

$$\begin{aligned}
 79. \int \frac{dx}{9+4x^2} &= \frac{1}{4} \int \frac{dx}{\frac{9}{4}+x^2} \\
 &= \frac{1}{4} \int \frac{dx}{\left(\frac{3}{2}\right)^2 + x^2} \\
 &= \frac{1}{4} \times \frac{2}{3} \tan^{-1} \frac{2}{3} x + c \\
 &= \frac{1}{6} \tan^{-1} \frac{2}{3} x + c \quad \therefore C
 \end{aligned}$$

$$\begin{aligned}
 80. \int_{\sqrt{2}}^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} dx &= \left[\sin^{-1} \frac{x}{2} \right]_{\sqrt{2}}^{\sqrt{3}} \\
 &= \left(\sin^{-1} \frac{\sqrt{3}}{2} \right) - \left(\sin^{-1} \frac{\sqrt{2}}{2} \right) \\
 &= \frac{\pi}{3} - \frac{\pi}{4} \\
 &= \frac{\pi}{12}
 \end{aligned}$$

$$\begin{aligned}
 81. \int_0^b \frac{dx}{b^2+x^2} &= \left[\frac{1}{b} \tan^{-1} \frac{x}{b} \right]_0^b \\
 &= \frac{1}{b} \left[\tan^{-1} \frac{b}{b} - \tan^{-1}(0) \right] \\
 &= \frac{1}{b} \times \tan^{-1}(1) \\
 &= \frac{1}{b} \times \frac{\pi}{4} = \frac{\pi}{3} \\
 \therefore b &= \frac{3}{4}
 \end{aligned}$$

$$\begin{aligned}
 82. \text{ Let } y &= x \cos^{-1} x - \sqrt{1-x^2} \\
 \frac{d}{dx} (x \cos^{-1} x) &= \cos^{-1} x + x \times \frac{-1}{\sqrt{1-x^2}} = \cos^{-1} x - \frac{x}{\sqrt{1-x^2}}
 \end{aligned}$$

$$\begin{aligned}
 \frac{d}{dx} \left((1-x^2)^{\frac{1}{2}} \right) &= \frac{1}{2} (1-x^2)^{-\frac{1}{2}} \cdot 2x = \frac{x}{\sqrt{1-x^2}} \\
 \therefore \frac{dy}{dx} &= \cos^{-1} x - \frac{x}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}} = \cos^{-1} x \\
 \therefore \int_0^1 \cos^{-1} x dx &= \left[x \cos^{-1} x - \sqrt{1-x^2} \right]_0^1 \\
 &= [\cos^{-1}(1) - (0-1)] \\
 &= 1
 \end{aligned}$$

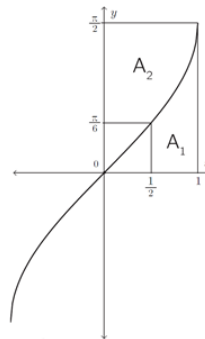
$$83. y = \sin^{-1} x$$

$$x = \sin y$$

$$\begin{aligned}
 \text{Area of rectangle} &= \frac{\pi}{6} \times \frac{1}{2} \\
 &= \frac{\pi}{12}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area} &= \frac{\pi}{12} - A_1 \\
 &= \frac{\pi}{12} - \int_0^{\pi/6} \sin y dy \\
 &= \frac{\pi}{12} + [\cos y]_0^{\pi/6} \\
 &= \frac{\pi}{12} + \left[\frac{\sqrt{3}}{2} - 1 \right] \\
 &= \frac{\pi}{12} + \frac{\sqrt{3}}{2} - 1 \text{ u}^2
 \end{aligned}$$

84.



$$\begin{aligned}
 \text{area } A_1 + A_2 &= \frac{\pi}{2} - \frac{\pi}{6} \times \frac{1}{2} \\
 &= \frac{\pi}{2} - \frac{\pi}{12} \\
 &= \frac{5\pi}{12}
 \end{aligned}$$

$$\therefore A_1 = \frac{5\pi}{12} - A_2$$

$$A_2 = \int_{\pi/6}^{\pi/2} \sin y dy$$

$$= - \left[\cos y \right]_{\pi/6}^{\pi/2}$$

$$= \left[\cos y \right]_{\pi/6}^{\pi/2}$$

$$= \frac{\sqrt{3}}{2} - 0$$

$$\therefore A_1 = \frac{5\pi}{12} - \frac{\sqrt{3}}{2} \text{ units}^2.$$

85.

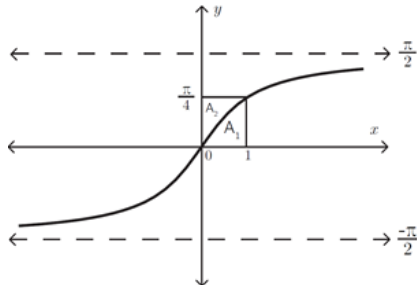
$$\text{i) } \int \tan x \, dx = \int \frac{\sin x}{\cos x} \, dx$$

$$u = \cos x \rightarrow \frac{du}{dx} = -\sin x$$

$$dx \sin x = -du$$

$$\begin{aligned} \int \tan x \, dx &= \int \frac{-du}{u} \\ &= -\ln(u) + C \\ &= -\ln(\cos x) + C \end{aligned}$$

ii)

Domain: all x Range: $-\frac{\pi}{2} < x < \frac{\pi}{2}$

$$\text{iii) } A_1 = \int_0^1 \tan^{-1} x \, dx = \frac{\pi}{4} \times 1 - A_2$$

$$\begin{aligned} A_2 &= \int_0^{\frac{\pi}{4}} \tan y \, dy \\ &= [-\ln(\cos y)]_0^{\frac{\pi}{4}} \quad (\text{from i}) \\ &= [\ln(\cos y)]_{\frac{\pi}{4}}^0 \\ &= \ln 1 - \ln \frac{1}{\sqrt{2}} \\ &= \frac{1}{2} \ln 2 \end{aligned}$$

$$\therefore \int_0^1 \tan^{-1} x \, dx = \frac{\pi}{4} - \frac{1}{2} \ln 2$$

Note: Extension 2 Students - In this question

integration by parts could have been an

alternative approach, however the question specifically stated that part (i) was to be used.

86.

$$\begin{aligned} \text{i) } g(x) &= \sin^{-1} x + \cos^{-1} x \\ g'(x) &= \frac{1}{\sqrt{1-x^2}} + \frac{-1}{\sqrt{1-x^2}} \\ &= 0 \end{aligned}$$

ii) If $g'(x) = 0$ then $g(x) = K$ for some constant K .Let $\sin^{-1} x + \cos^{-1} x \equiv K$. Pick $x = \frac{1}{2}$,

$$\begin{aligned} LHS &= \sin^{-1}\left(\frac{1}{2}\right) + \cos^{-1}\left(\frac{1}{2}\right) \\ &= \frac{\pi}{6} + \frac{\pi}{3} \\ &= \frac{\pi}{2} \\ \therefore K &= \frac{\pi}{2} \end{aligned}$$

$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

$$\begin{aligned} \therefore \int_0^1 \sin^{-1} x + \cos^{-1} x \, dx &= \int_0^1 \frac{\pi}{2} \, dx \\ &= \left[\frac{\pi}{2} x \right]_0^1 \\ &= \frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} \text{87. } \int_0^k \frac{1}{\sqrt{4-x^2}} \, dx &= \left[\sin^{-1}\left(\frac{x}{2}\right) \right]_0^k = \frac{\pi}{3} \\ &= \left[\sin^{-1}\left(\frac{k}{2}\right) - \sin^{-1}(0) \right] = \frac{\pi}{3} \\ \therefore \sin^{-1}\left(\frac{k}{2}\right) &= \frac{\pi}{3} \\ \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) &= \frac{\pi}{3} \\ \therefore k &= \sqrt{3} \\ \therefore B \end{aligned}$$

$$\begin{aligned} \text{88. } I &= \int \frac{1}{1+4x^2} \, dx \\ &= \int \frac{1}{4\left(\frac{1}{4}+x^2\right)} \, dx \\ &= \frac{1}{4} \int \frac{1}{\left(\frac{1}{2}\right)^2+x^2} \, dx \\ &= \frac{1}{4} \times 2 \tan^{-1} 2x + C \\ &= \frac{1}{2} \tan^{-1} 2x + c \quad \therefore D \end{aligned}$$

$$\begin{aligned} \text{89. } \int_0^{\sqrt{3}} \frac{1}{\sqrt{4-x^2}} \, dx &= \left[\sin^{-1} \frac{x}{2} \right]_0^{\sqrt{3}} \\ &= \left[\sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} 0 \right] \\ &= \frac{\pi}{3} \end{aligned}$$

$$\begin{aligned} \text{90. } \int \frac{1}{\sqrt{4-9x^2}} \, dx &= \int \frac{1}{3\sqrt{\frac{4}{9}-x^2}} \, dx \\ &= \frac{1}{3} \int \frac{1}{\sqrt{\left(\frac{2}{3}\right)^2-x^2}} \, dx \\ &= \frac{1}{3} \sin^{-1}\left(\frac{3x}{2}\right) + C. \end{aligned}$$

Volumes of Solids of Revolution

$$\begin{aligned}
 91. \text{ Volume} &= \pi \int_1^2 x^2 dy \\
 y &= e^{x/2} \\
 \therefore \frac{x}{2} &= \ln y \\
 \therefore x &= 2 \ln y
 \end{aligned}$$

$$\begin{aligned}
 \text{Volume} &= \pi \int_1^2 (2 \ln y)^2 dy \\
 &= 4\pi \int_1^2 (\ln y)^2 dy \therefore D
 \end{aligned}$$

$$\begin{aligned}
 92. V &= \pi \int_1^4 \left(\frac{1}{x^2}\right)^2 dx = \pi \int_1^4 x^{-4} dx \\
 &= \pi \left[\frac{x^{-3}}{-3} \right]_1^4 \\
 &= -\frac{\pi}{3} \left[\frac{1}{x^3} \right]_1^4 \\
 &= \frac{\pi}{3} \left[\frac{1}{x^3} \right]_4^1 \\
 &= \frac{\pi}{3} \left[1 - \frac{1}{64} \right] \\
 &= \frac{21\pi}{64} u^3
 \end{aligned}$$

$$\begin{aligned}
 93. y &= \frac{1}{x} + 1 \\
 \frac{1}{x} &= y - 1 \rightarrow x = \frac{1}{y-1} \\
 V &= \pi \int_2^4 \frac{1}{(y-1)^2} dy \\
 V &= \pi \int_2^4 (y-1)^{-2} dy \\
 &= \pi \left[\frac{(y-1)^{-1}}{-1} \right]_2^4 \\
 &= -\pi \left[\frac{1}{y-1} \right]_2^4 \\
 &= \pi \left[\frac{1}{y-1} \right]_4^2 \\
 &= \pi \left[1 - \frac{1}{3} \right] = \frac{2\pi}{3} u^3
 \end{aligned}$$

$$\begin{aligned}
 94. V &= \pi \int_1^2 (e^{-x})^2 dx = \pi \int_1^2 e^{-2x} dx \\
 &= \pi \left[-\frac{1}{2} e^{-2x} \right]_1^2 \\
 &= \frac{\pi}{2} [e^{-2x}]_2^1 \\
 &= \frac{\pi}{2} [e^{-2} - e^{-4}] \\
 &= \frac{\pi}{2} \left[\frac{1}{e^2} - \frac{1}{e^4} \right] u^3
 \end{aligned}$$

$$95. y = \tan^{-1} x \rightarrow x = \tan y$$

$$\begin{aligned}
 V &= \pi \int_0^{\frac{\pi}{4}} \tan^2 y dy = \pi \int_0^{\frac{\pi}{4}} \sec^2 y - 1 dy \\
 &= \pi [\tan y - y]_0^{\frac{\pi}{4}} \\
 &= \pi \left[\left(\tan \frac{\pi}{4} - \frac{\pi}{4} \right) - 0 \right] \\
 &= \pi \left(1 - \frac{\pi}{4} \right) \\
 &= \frac{\pi}{4} (4 - \pi) u^3
 \end{aligned}$$

96. Volume generated by the line

$$\begin{aligned}
 V_1 &= \pi \int_{\frac{1}{2}}^2 \left(-x + 2\frac{1}{2} \right)^2 dx \\
 &= \pi \left[\frac{(-x + 2\frac{1}{2})^3}{3} \times -1 \right]_{\frac{1}{2}}^2 \\
 &= -\frac{\pi}{3} \left[\left(-x + 2\frac{1}{2} \right)^3 \right]_{\frac{1}{2}}^2 \\
 &= \frac{\pi}{3} \left[\left(-x + 2\frac{1}{2} \right)^3 \right]_{\frac{1}{2}}^{1/2} = \frac{\pi}{3} \left[8 - \frac{1}{8} \right] = \frac{21\pi}{8} u^3
 \end{aligned}$$

Volume generated by hyperbola

$$\begin{aligned}
 V_2 &= \pi \int_{\frac{1}{2}}^2 \frac{1}{x^2} dx \\
 &= \pi \int_{\frac{1}{2}}^2 x^{-2} dx \\
 &= \pi \left[\frac{x^{-1}}{-1} \right]_{\frac{1}{2}}^2 \\
 &= \pi \left[\frac{1}{x} \right]_{\frac{1}{2}}^{1/2} = \pi \left[2 - \frac{1}{2} \right] = \frac{3\pi}{2}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Volume} &= V_1 - V_2 \\
 &= \frac{21\pi}{8} - \frac{3\pi}{2} \\
 &= \frac{9\pi}{8} u^3
 \end{aligned}$$

$$\begin{aligned}
 97. V &= \pi \int_a^b y^2 dx \\
 &= \pi \int_{-\pi}^{\pi} 1 - \cos x dx \\
 &= \pi [x - \sin x]_{-\pi}^{\pi} \\
 &= \pi [(\pi - \sin \pi) - (-\pi - \sin(-\pi))] \\
 &= \pi [(\pi - 0) - (-\pi - 0)] \\
 &= 2\pi^2 \text{ units}^3
 \end{aligned}$$

$$98. y = x^2 - 2 \rightarrow x^2 = y + 2$$

$$\begin{aligned}
 V &= \pi \int_{-2}^0 y + 2 dy = \pi \left[\frac{y^2}{2} + 2y \right]_{-2}^0 \\
 &= \pi [0 - (2 - 4)] \\
 &= 2\pi u^3
 \end{aligned}$$

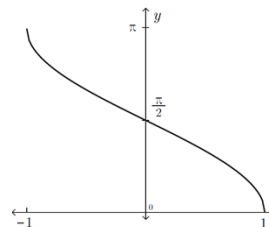
99. For $x = y^2$: $V_1 = \pi \int_0^1 x dx = \pi \left[\frac{x^2}{2} \right]_0^1 = \frac{\pi}{2}$
 For $y = x^2$: $V_2 = \pi \int_0^1 x^4 dx = \pi \left[\frac{x^5}{5} \right]_0^1 = \frac{\pi}{5}$
 $\therefore \text{Volume} = V_1 - V_2 = \frac{\pi}{2} - \frac{\pi}{5} = \frac{3\pi}{10} \text{ u}^3$

100. $y = x^2 - 2$
 $x^2 = y + 2$
 $V = \pi \int_1^4 x^2 dy$
 $= \pi \int_1^4 y + 2 dy$
 $= \pi \left[\frac{y^2}{2} + 2y \right]_1^4$
 $= \pi \left[(8 + 8) - \left(\frac{1}{2} + 2 \right) \right]$
 $= 13.5 \pi \text{ u}^3$

101. Let $y = 0$ to find the x -intercepts,
 $x^2 = 12 \Rightarrow x = \pm \sqrt{12}$
 $V = \pi \int y^2 dy$
 $x^2 + 3y^2 = 12$
 $\therefore y^2 = \frac{12 - x^2}{3}$
 $V = \pi \int_{-\sqrt{12}}^{\sqrt{12}} \frac{12 - x^2}{3} dx$
 $= \frac{2\pi}{3} \int_0^{\sqrt{12}} 12 - x^2 dx \quad (\text{even function})$
 $= \frac{2\pi}{3} \left[12x - \frac{x^3}{3} \right]_0^{\sqrt{12}}$
 $= \frac{2\pi}{3} \left[12\sqrt{12} - \frac{12\sqrt{12}}{3} \right]$
 $= \frac{32\pi\sqrt{3}}{3} \text{ units}^3$

102.

i) = Domain: $-1 \leq x \leq 1$. Range: $0 \leq y \leq \pi$.



ii) $V = \pi \int_0^{\frac{\pi}{2}} x^2 dy$
 $x = \cos y$
 $V = \pi \int_0^{\frac{\pi}{2}} \cos^2 y dy$
 $= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} 1 + \cos 2y dy$
 $= \frac{\pi}{2} \left[y + \frac{1}{2} \sin 2y \right]_0^{\frac{\pi}{2}}$
 $V = \frac{\pi}{2} \left[\left(\frac{\pi}{2} + 0 \right) - 0 \right]$
 $= \frac{\pi^2}{4} \text{ units}^3$

103.

$$\begin{aligned} \text{i) } \int_0^{\frac{3\pi}{2}} \sin^2 x \, dx &= \frac{1}{2} \int_0^{\frac{3\pi}{2}} 1 - \cos 2x \, dx \\ &= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{3\pi}{2}} \\ &= \frac{1}{2} \left[\left(\frac{3\pi}{2} - 0 \right) - 0 \right] \\ &= \frac{3\pi}{4} \end{aligned}$$

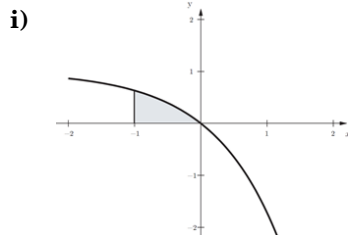
$$\begin{aligned} \text{ii) } V &= \pi \int_0^{\frac{3\pi}{2}} (1 + \sin x)^2 \, dx \\ &= \pi \int_0^{\frac{3\pi}{2}} 1 + 2\sin x + \sin^2 x \, dx \\ &= \pi \int_0^{\frac{3\pi}{2}} 1 + 2\sin x \, dx + \pi \int_0^{\frac{3\pi}{2}} \sin^2 x \, dx \\ &= \pi \left[x - 2\cos x \right]_0^{\frac{3\pi}{2}} + \frac{3\pi^2}{4} \\ &= \pi \left[\left(\frac{3\pi}{2} - 0 \right) - (-2) \right] + \frac{3\pi^2}{4} \\ &= \frac{3\pi^2}{2} + 2\pi + \frac{3\pi^2}{4} \\ &= \frac{9\pi^2}{4} + 2\pi \text{ units}^3. \end{aligned}$$

104.

$$\begin{aligned} \text{i) } V &= \pi \int x^2 \, dy \\ y &= \ln x \rightarrow x = e^y \\ \text{when } x &= 4, y = \ln 4 \\ \therefore V &= \pi \int_0^{\ln 4} (e^y)^2 \, dy \\ V &= \pi \int_0^{\ln 4} e^{2y} \, dy \end{aligned}$$

$$\begin{aligned} \text{ii) } V &= \pi \int_0^{\ln 4} e^{2y} \, dy \\ &= \pi \left[\frac{1}{2} e^{2y} \right]_0^{\ln 4} \\ &= \frac{\pi}{2} [e^{2\ln 4} - e^0] \\ &= \frac{\pi}{2} [e^{\ln 16} - 1] \\ &= \frac{\pi}{2} \times (16 - 1) \\ &= \frac{15\pi}{2} \text{ units}^3 \end{aligned}$$

105.



$$\begin{aligned} \text{ii) } V &= \pi \int_{-1}^0 (1 - e^x)^2 \, dx \\ &= \pi \int_{-1}^0 1 - 2e^x + e^{2x} \, dx \\ &= \pi \left[x - 2e^x + \frac{1}{2} e^{2x} \right]_{-1}^0 \\ &= \pi \left[\left(0 - 2 + \frac{1}{2} \right) - \left(-1 - 2e^{-1} + \frac{1}{2} e^{-2} \right) \right] \\ &= \pi \left[-\frac{1}{2} + 2e^{-1} - \frac{1}{2} e^{-2} \right] \\ &\approx 0.528074 \text{ units}^3 \end{aligned}$$

106.

$$\begin{aligned} \text{i) } V &= \pi \int_0^{\frac{\pi}{2}} \left(\tan \frac{x}{2} \right)^2 \, dx \\ &= \pi \int_0^{\frac{\pi}{2}} \left(\sec^2 \frac{x}{2} - 1 \right) \, dx \quad \left(1 + \tan^2 x = \sec^2 x \right) \\ \text{ii) } V &= \pi \int_0^{\frac{\pi}{2}} \left(\sec^2 \frac{x}{2} - 1 \right) \, dx \\ &= \pi \left[2 \tan \frac{x}{2} - x \right]_0^{\frac{\pi}{2}} \\ &= \pi \left[\left(2 \tan \frac{\pi}{4} - \frac{\pi}{2} \right) - 0 \right] \\ &= \pi \left[2 - \frac{\pi}{2} \right] \\ &= \frac{\pi}{2} (4 - \pi) \text{ units}^3 \end{aligned}$$

107. Point of intersection: $\sin x = \cos 2x$

$$\begin{aligned} \sin x &= 1 - 2\sin^2 x \\ 2\sin^2 x + \sin x - 1 &= 0 \\ (2\sin x - 1)(\sin x + 1) &= 0 \\ \sin x &= 1/2 \text{ or } \sin x = -1 \\ \therefore x &= \frac{\pi}{6} \end{aligned}$$

$$\begin{aligned} V &= \pi \int_0^{\frac{\pi}{6}} \cos^2 2x \, dx - \pi \int_0^{\frac{\pi}{6}} \sin^2 x \, dx \\ &= \frac{\pi}{2} \int_0^{\frac{\pi}{6}} \cos 4x + 1 \, dx - \frac{\pi}{2} \int_0^{\frac{\pi}{6}} 1 - \cos 2x \, dx \\ &= \frac{\pi}{2} \left[\frac{1}{4} \sin 4x + x \right]_0^{\frac{\pi}{6}} - \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{6}} \\ &= \frac{\pi}{2} \left[\frac{1}{4} \sin \frac{2\pi}{3} + \frac{\pi}{6} - 0 \right] - \frac{\pi}{2} \left[\frac{\pi}{6} - \frac{1}{2} \sin \frac{\pi}{3} - 0 \right] \\ &= \frac{\pi}{2} \left[\frac{\sqrt{3}}{8} + \frac{\pi}{6} \right] - \frac{\pi}{2} \left[\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right] \\ &= \frac{\pi}{2} \left[\frac{\sqrt{3}}{8} + \frac{\pi}{6} - \frac{\pi}{6} + \frac{\sqrt{3}}{4} \right] \\ &= \frac{\pi}{2} \times \frac{3\sqrt{3}}{8} \\ &= \frac{3\sqrt{3}\pi}{16} \text{ units}^3 \end{aligned}$$

108. $V = \pi \int x^2 dy$ about the y -axis

$$\begin{aligned}\therefore V &= \pi \int_0^2 y dy - \pi \int_1^2 y - 1 dy \\ &= \pi \left[\frac{y^2}{2} \right]_0^2 - \pi \left[\frac{y^2}{2} - y \right]_1^2 \\ &= 2\pi - \pi \left[(2-2) - \left(\frac{1}{2} - 1 \right) \right] \\ &= 2\pi - \frac{\pi}{2} \\ &= \frac{3\pi}{2} m^3\end{aligned}$$

109. $V = \pi \int_0^{\frac{\pi}{2k}} (k+1)^2 \sin^2 kx dx$

$$= \pi(k+1)^2 \int_0^{\frac{\pi}{2k}} \sin^2 kx dx$$

Now $\cos 2x = 1 - 2\sin^2 x \rightarrow \sin^2 x = \frac{1}{2}(1 - \cos 2x)$

$$\begin{aligned}\therefore V &= \frac{\pi(k+1)^2}{2} \int_0^{\frac{\pi}{2k}} 1 - \cos 2kx dx \\ &= \frac{\pi(k+1)^2}{2} \left[x - \frac{1}{2} \sin 2kx \right]_0^{\frac{\pi}{2k}} \\ &= \frac{\pi(k+1)^2}{2} \left[\frac{\pi}{2k} - \frac{1}{2} \sin \pi - 0 \right]\end{aligned}$$

$$\begin{aligned}V &= \pi^2 \rightarrow \therefore \frac{\pi(k+1)^2}{2} \left[\frac{\pi}{2k} \right] = \pi^2 \\ (k+1)^2 &= 4k \\ k^2 + 2k + 1 &= 4k \\ k^2 - 2k + 1 &= 0 \\ (k-1)^2 &= 0 \\ \therefore k &= 1\end{aligned}$$

110. $V = \pi \int_4^{12} x^2 dy + \pi \times 10^2 \times 4$

$$\begin{aligned}&= \pi \int_4^{12} \left(\frac{60}{y} - 5 \right)^2 dy + 400\pi \\ &= \pi \int_4^{12} 3600y^{-2} - \frac{600}{y} + 25 dy + 400\pi \\ &= \pi \left[-\frac{3600}{y} - 600 \ln y + 25y \right]_4^{12} + 400\pi \\ &= \pi [(-300 - 600 \ln 12 + 300) - (-900 - 600 \ln 4 + 100)] + 400\pi \\ &= \pi [800 - 600 \ln 3] + 400\pi \\ &= 800\pi - 600\pi \ln 3 + 400\pi \\ &= 1200\pi - 600\pi \ln 3\end{aligned}$$